Uniform Bessel bounds for magnetization crossings in finite Ising chains

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Abstract

We consider a finite Ising chain in zero field and ask when the all-up configuration is as likely as zero magnetization (magnetization one for an odd number of spins). For both free and periodic ends, the crossing coupling of a long chain is half the logarithm of its length minus half the logarithm of that logarithm, up to a term that vanishes as the length grows. We compute further terms with explicit constants and, for free ends, bound the crossing at every length. The second term is missing from an earlier approximation and explains why an earlier numerical fit gives a slope below one half. To locate the crossing, we compare the known magnetization law with its Bessel approximation. The Bessel approximation is above the exact weight, and we bound its relative error for every length, coupling and magnetization. At the crossing this bound tends to zero. The central bin still has about as many domain walls as the logarithm of the length on average, so the crossing has a coupling width of order the inverse logarithm. In a fixed positive dimensionless field the last crossing comes from near the edge of the row, at a coupling of about a quarter of the logarithm of the length. For long chains it is the one-spin crossing when the field exceeds half the logarithm of two. Below that value another bin crosses later.

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1 Introduction

1.1 The crossing

Let $Z_N(M)$ be the sum of the Ising weights of the configurations of N spins with magnetization M (the partition function of the fixed-magnetization ensemble). Counting configurations by their domain walls gives a classical finite sum [1, 2, 3]. When the mean number of walls stays bounded as N grows, this sum has a modified Bessel approximation [2, 4, 5]. We bound its relative error explicitly for every length, coupling and magnetization (Theorem 4.3 and Corollary 4.4 for free ends, Theorem 5.1 for periodic ends). The Bessel form always overestimates the exact weight. Its error also stays small when the mean number of walls grows like $\log N$, which is the regime of the crossing we study.

Take $N \geq 2$ spins $\sigma_i = \pm 1$ with free ends and zero-field Ising weight $\exp(J \sum_{i<N} \sigma_i \sigma_{i+1})$, where $J > 0$. Here J is dimensionless, and it is the product βJ_{ex} of the inverse temperature and the exchange coupling J_{ex} of [6]. The all-up configuration is the only one with magnetization $\sigma_1 + \cdots + \sigma_N = N$. Zero magnetization (or magnetization ± 1 for odd N) comes from many

configurations. For small J domain walls cost little and a central magnetization is the more likely of the two, and for large J the all-up configuration is. They are equally likely at a single coupling, the *central crossing*. Stepanyan et al. [6] study this crossing in the periodic chain as a thermal transition of a finite system (the analogue of the transition temperature of a first-order transition). They approximate it by $\beta J \approx (2\log N - 1)/5$. Like them, we compare one central bin with the single all-up configuration (Remark 5.6 says what changes for other comparisons). We prove the asymptotics of the crossing with explicit constants for free and for periodic ends, and for free ends we bound it at every N.

Read from left to right, the spins are the steps of a persistent random walk [7, 8], a correspondence that García-Pelayo [9] uses to derive the exact magnetization laws of the free and the periodic chain, also in a field. The first step is fair, and each later step repeats the previous one with probability p, where $e^{-2J} = (1-p)/p$. The magnetization is the position after N steps, so the walk that never turns, which reaches an end, competes with the many walks that end in the middle.

With two steps the walk ends in the middle exactly when it turns once, so the three bins have probabilities $\frac{p}{2}$, $1-p$ and $\frac{p}{2}$, and they are level at $p = \frac{2}{3}$. With three steps the bins have probabilities $\frac{p^2}{2}$, $\frac{1-p^2}{2}$, $\frac{1-p^2}{2}$ and $\frac{p^2}{2}$, and they are level at $p = 1/\sqrt{2}$. In terms of the coupling these two crossings are at $J = \frac{1}{2}\log 2$ and $J = \frac{1}{2}\log(1+\sqrt{2})$. In general exactly one walk reaches a given endpoint, namely the walk that never turns, and very many walks reach the middle by turning a few times (each turn brings a factor of about $1-p$).

The middle bin and the endpoints are level when the walk turns about $\log N$ times on average. More precisely, the crossing value p_N of p satisfies $N(1-p_N) = \log N - \frac{1}{2}\log\log N - 0.467\ldots + o(1)$, and for every $N \geq 2$ we trap $N(1-p_N)$ between two explicit functions of N whose distance tends to 0 (Theorem 3.4). For $p < p_N$ the middle is more likely, and for $p > p_N$ the ends are. In terms of the coupling, the crossing is at $J = \frac{1}{2}(\log N - \log\log N) + O(\log\log N/\log N)$ for free and for periodic ends, and the correlation length there is about $N/(2\log N)$ (Corollary 5.5). Their approximation and their fitted line [6] have no $\log\log N$ term, and this term explains why the slope of the fit is below $\frac{1}{2}$ (Section 5.3). At the free crossing the central bin carries about $\log N$ domain walls, so its mean energy exceeds that of the all-up configuration by only $2\log N - \log\log N + O(1)$ (Proposition 5.7). The crossing therefore has a width of order $1/\log N$ in the coupling, not $1/N$ as at a first-order transition.

The crossing lies between two regimes in which the law is well understood. When $p = 1 - \tau/N$ with $\tau > 0$ fixed, the rescaled walk tends to the integrated telegraph process and the magnetization per spin has a scaling limit [10, Theorem 2.1 and Remark 3.4]. When $0 < p < 1$ is fixed, the magnetization has a normal limit (Theorem 2.5). At the crossing τ grows like $\log N$, so neither limit decides it, and we need bounds that hold uniformly in τ. Our main tool is a comparison of the middle bin with a modified Bessel function, with a relative error that is explicit for every N (Theorem 3.3). A walk with d directions, which is the d-state Potts chain, is treated in [11].

1.2 Setting and notation

The walk. The walk makes $N \geq 1$ steps. We record a path of the walk as a word $w = w_1 \cdots w_N$ in the letters $+$ (right) and $-$ (left). The letter w_1 is $+$ or $-$ with probability $\frac{1}{2}$, and given $w_1, \ldots, w_i$,

$$\mathbb{P}(w_{i+1} = w_i) = p, \qquad \mathbb{P}(w_{i+1} \neq w_i) = q = 1 - p.$$

An index $i < N$ with $w_i \neq w_{i+1}$ is a *switch*. Let $\mathrm{chg}(w)$ be the number of switches and $\mathrm{rep}(w) = N - 1 - \mathrm{chg}(w)$. Then

$$\mathbb{P}(w) = \tfrac{1}{2}\, p^{\mathrm{rep}(w)} q^{\mathrm{chg}(w)}. \tag{1}$$

For $p > 0$ we put $u = q/p$, the odds of a switch.

Bins. The walk ends in bin k if w has k letters $+$, for $0 \le k \le N$. We write $P_N(k)$ for the probability of bin k, and we set $P_0(0) = 1$. We call the bins $0, \dots, N$ together the *row* (as in a Galton board), and bins 0 and N are its *endpoints*. Only the constant word $-\cdots-$ reaches bin 0, and only $+\cdots+$ reaches bin N, so

$$P_N(0) = P_N(N) = \tfrac{1}{2} p^{N-1} \qquad (N \ge 1).$$

We call bin $m_N = \lfloor N/2 \rfloor$ a middle bin. For odd N the two middle bins m_N and $m_N + 1$ have equal probability by symmetry.

Spins. Put $\sigma_i = +1$ if $w_i = +$ and $\sigma_i = -1$ if $w_i = -$, and $X_N = \sigma_1 + \cdots + \sigma_N$. The bin is $K_N = (N + X_N)/2$. Since $\mathrm{chg}(w) = \sum_{i<N}(1 - \sigma_i\sigma_{i+1})/2$, (1) shows that $\mathbb{P}(w)$ is proportional to $\exp(J \sum_{i<N} \sigma_i\sigma_{i+1})$ with $e^{-2J} = u$. So the spins form the Ising chain in zero field with free ends, and $J > 0$ exactly when $p > \frac{1}{2}$.

Scales. Let $L = \log N$ and $\tau = Nq$. Each of the $N - 1$ steps after the first switches with probability q, so the expected number of switches is $(N-1)q = \tau - q$. We call τ the *switching scale*. In the Ising chain a switch is a domain wall, and $q = 1/(1 + e^{2J})$, so τ is the expected number of domain walls plus q. The central crossing has τ of order L. Near the edges the crossings have larger τ, up to order $\sqrt{N}$ for the bins next to an edge (Section 4). We use several equivalent parameters, namely the persistence p, the odds $u = q/p = e^{-2J}$ of a switch, the coupling J, the switching scale $\tau = Nq$, and in the Bessel estimates $w = Nu$ and $x = w/2$ (from Section 3 on, the context distinguishes it from a word w). Near the central crossing u is of order L/N, and then τ and $Nu = 2x$ differ by $O(L^2/N)$.

Ratios and crossings. For a bin k with $1 \le k \le N - 1$ we write $a = k$ and $b = N - k$ for its numbers of letters $+$ and $-$. For $p > 0$ the ratio of bin k to an endpoint depends on p only through u, and it is a polynomial in u with nonnegative integer coefficients (see (13) below). We write

$$S_{a,b}(u) = \frac{P_N(a)}{P_N(0)}, \qquad S_N(u) = S_{m_N,\, N-m_N}(u)$$

for this ratio and for its value at the middle. *The crossing* of bin k is the value of p at which bin k is exactly as likely as an endpoint, that is, $S_{a,b}(u) = 1$. It is unique (Section 4.1), and we write $p_{a,b}$ for it and $u_{a,b} = (1 - p_{a,b})/p_{a,b}$ for its odds. The *central crossing* is $p_N = p_{m_N, N-m_N}$, with odds u_N, and

$$z_N = N(1 - p_N) = \frac{N u_N}{1 + u_N} \tag{2}$$

is the value of τ there.

Special functions and asymptotic notation. The functions I_0 and I_1 are the modified Bessel functions of the first kind, of orders zero and one. We use them through their power series and their behavior at infinity [12, Eqs. 10.25.2, 10.40.1],

$$I_0(2x) = \sum_{r\ge 0} \frac{x^{2r}}{(r!)^2}, \quad I_1(2x) = \sum_{r\ge 0} \frac{x^{2r+1}}{r!\,(r+1)!}, \quad I_\nu(z) = \frac{e^z}{\sqrt{2\pi z}}\Big(1 - \frac{4\nu^2-1}{8z} + O(z^{-2})\Big)$$

as $z \to +\infty$ along the positive real axis, for $\nu = 0, 1$. We write $f \sim g$ if $f/g \to 1$. Each $O(\cdot)$ comes with the parameters it is uniform in, and $O_a(\cdot)$ means that the constant may depend on a.

Ising dictionary. Sections 2 to 4 are written in the variables of the walk, and they translate as follows. Bin k is magnetization $X_N = 2k - N$, so the endpoints are full magnetization $\pm N$. A middle bin has magnetization zero for even N and either -1 or 1 for odd N. A switch is a domain wall, and its odds $u = e^{-2J}$ are the Boltzmann factor of one wall. For free ends, $Z_N(M)$ is the sum of the weights $\exp(J\sum_{i<N}\sigma_i\sigma_{i+1})$ of the configurations with magnetization M, so $Z_N(N) = e^{(N-1)J}$ and $S_{a,b}(u) = Z_N(a-b)/Z_N(N)$. The crossing of bin k is the coupling $-\frac{1}{2}\log u_{a,b}$ at which magnetization $a-b$ and magnetization N are equally likely, and at the center it is $J_N^{\mathrm{F}} = -\frac{1}{2}\log u_N$. The correlation length of the infinite chain in zero field, with no constraint on the magnetization, is $\xi(J) = -1/\log\tanh J$, and $\tanh J = (1-u)/(1+u)$, so $N/\xi = N\log\frac{1+u}{1-u} = 2Nu(1+O(u^2))$. Since $\tau = Nu/(1+u)$, this gives $\tau = \frac{N}{2\xi}(1+O(u))$ as $u \to 0$. To leading order, the switching scale is half the length of the chain measured in correlation lengths. It is exactly the scaling variable $\zeta = N(1-\tanh J)/2$ of Antal, Droz and Rácz [2] (we keep the letter ζ for ζ_N below).

1.3 Main results

The coupling at the crossing. Let J_N^{P} be the coupling at the periodic central crossing, and J_N^{F} the free one of Section 1.2. Put $c_0 = \frac{1}{2}\log(\pi/8) = -0.467\ldots$ and $c_{\mathrm{P}} = \frac{1}{2}\log(\pi/2)$. Theorem 5.2 and Corollary 5.5 give

$$J_N^X = \tfrac{1}{2}(\log N - \log\log N) + \frac{\frac{1}{4}\log\log N - \frac{1}{2}c_X}{\log N} + O\Big(\frac{(\log\log N)^2}{(\log N)^2}\Big)$$

for free ends ($X = \mathrm{F}$, with $c_{\mathrm{F}} = c_0$) and for periodic ends ($X = \mathrm{P}$). Since $c_{\mathrm{P}} = c_0 + \log 2$, periodic ends lower the coupling at the crossing by $(\log 2)/(2\log N)$ to leading order. At either crossing the correlation length is about $N/(2\log N)$, a vanishing fraction of the chain. The approximation $\beta J \approx (2\log N - 1)/5$ of [6, Eq. (16)] grows too slowly. At $N = 10^5$ the exact periodic coupling is 4.576. The periodic case of the display without its error term gives 4.578, and the approximation gives 4.405 (Section 5.3).

A Bessel bound at every N. Put $h = N/2$ and $F(x) = 2x(I_0(2x) + I_1(2x))$. Theorem 3.3 says that for every $N \ge 2$ and every $x > 0$,

$$0 \le 1 - \frac{hS_N(x/h)}{F(x)} \le \frac{x(x+1)}{h}.$$

Near the crossing $x = Nu/2$ is about $L/2$, so the relative error is $O(L^2/N)$. Theorem 4.3 gives the same kind of bound for every bin, with relative error at most $Nu^2/2 + u$.

The central crossing at every N. For free ends, let ζ_N be the positive root of $8\zeta e^{2\zeta} = \pi N^2$, that is, $\zeta_N = \frac{1}{2}W_0(\pi N^2/4)$ with W_0 the principal branch of the Lambert function, and let $\varepsilon_N = \zeta_N(\zeta_N+1)/N + 1/(16\zeta_N^2)$. Theorem 3.4 says that

$$\zeta_N + \frac{1}{8\zeta_N} - \varepsilon_N < z_N < \zeta_N + \frac{1}{8\zeta_N} \qquad \text{for every } N \geq 2.$$

Note that $\varepsilon_N \to 0$. In terms of the coupling, with $Y_N = \zeta_N + \frac{1}{8\zeta_N}$ and $Y'_N = Y_N - \varepsilon_N$, this gives $\frac{1}{2}\log(N/Y_N - 1) < J_N^{\mathrm{F}} < \frac{1}{2}\log(N/Y'_N - 1)$ (Corollary 3.7(a)). For $N \geq 175$ the proof is analytic. For $2 \leq N \leq 174$ we decide the sign of $S_N - 1$ at two rationals, one on each side of the root, in exact integer arithmetic, compare the resulting bracket with ζ_N in outward interval arithmetic, and check both steps again in the Lean proof assistant. The same bounds give two simpler statements. With c_0 as above, the crossing satisfies

$$z_N + \tfrac{1}{2}\log z_N = L + c_0 + \frac{1}{8z_N} + R_N, \qquad |R_N| \leq \frac{1}{4L^2} + \frac{L^2+L}{N} \quad (N \geq 2)$$

(Proposition 3.8), and as $N \to \infty$

$$z_N = L - \tfrac{1}{2}\log L + c_0 + \frac{\frac{1}{4}\log L - \frac{1}{2}c_0 + \frac{1}{8}}{L} + O\Big(\frac{(\log L)^2}{L^2}\Big)$$

(Corollary 3.7(c)). The convergence is slow, and z_N/L is still only 0.84 at $N = 10^4$ (Section 3).

Every bin. For free ends, let bin a be at distance a from the nearer edge, with $b = N - a \geq a$. Next to the edge the crossing is at $p_N^{\mathrm{adj}} = (1+\sqrt{N-1})/(2+\sqrt{N-1})$. For $N \geq 4$ the two bins next to the edges are the least likely of all at the central crossing (Proposition 4.2). As $a \to \infty$, Theorem 4.9 expands $2\sqrt{ab}\,u_{a,b}$ to the same order as z_N, uniformly in $b \geq a$. For fixed a the crossing comes instead from a Laguerre polynomial, and $u_{a,b} = c_a/\sqrt{b} - 1/b + O_a(b^{-3/2})$ as $b \to \infty$, where c_a^2 is the positive root of an explicit polynomial of degree a (Theorem 4.10). The two regimes match (Theorem 4.11).

Periodic boundaries. If the steps are closed into a loop, so that the last step also interacts with the first, the central crossing moves by $(\log 2)/N$ in the odds to leading order (Theorem 5.2). Under either boundary, at odds $u_N^X + t/N$, where u_N^X is the central crossing of that boundary, the ratio of the bin nearest to $N/2 + yN/\sqrt{\log N}$ to an endpoint tends to e^{t-2y^2} as $N \to \infty$, for fixed t and y (Theorem 5.3).

Domain walls and the energy gap. At the free central crossing, the central bin has $z_N + \frac{1}{2} + O(1/z_N)$ domain walls on average, so its mean energy exceeds that of the all-up configuration by $2\log N - \log\log N + O(1)$ in units of the exchange coupling (Proposition 5.7). Hence a factor e change in their probability ratio takes a coupling change of about $1/(2\log N)$, whose slow decay we discuss in Section 6.

A magnetic field. Fix a dimensionless field $\mathcal{B} = \beta H > 0$ (written βh in [6]). For free and for periodic ends, both comparisons behind the transitions of [6] then come from the edge of the row, at $J = \frac{1}{4}\log N + O(1)$, and we find the limit of the constant. For large N and $\mathcal{B} > \mathcal{B}_c = \frac{1}{2}\log 2$, the one-spin crossing is the last crossing of the all-up configuration as J increases. Below $\mathcal{B}_c$

another bin crosses later, and at $\mathcal{B}_c$ the two gap between the two crossing couplings tends to zero (Theorem 5.9).

We prove these results in Sections 2 to 5 and discuss them in Section 6. Section 7 lists what was computed and what is in Lean, and Appendix A has the longer proofs of Section 3.

1.4 Related work

Early treatments of the persistent (or correlated) random walk include Goldstein [7] and Gillis [8]. Renshaw and Henderson [13] study the symmetric correlated walk, derive exact transition probabilities and investigate limiting distributions. Konno [14, Lemma 2 and Theorem 7] writes the exact law with the hypergeometric function ${}_2F_1$ and proves the integrated telegraph limit when $p = 1 - \theta/N$ with $\theta \in (0, 1)$ fixed. Herrmann and Vallois [10, Theorem 2.1 and Remark 3.4] give the process limit and its law, including the Bessel density and endpoint atoms. With τ replaced by Nu, their law gives the Bessel form of every bin that we use in Section 4.2. These fixed-scale limits do not by themselves give the uniform finite-N bounds needed when τ grows like $\log N$. Dekking and Kong [15] study the Markov binomial law, give its variance and find when it has one, two or three modes. Their results already imply the crossing of the bins next to the edges and the ordering of the bin probabilities within a row (Section 4.1).

For the Ising chain, Antal, Droz and Rácz [2] give the magnetization law at a fixed switching scale τ under several boundary conditions. Dantchev and Rudnick [3] give the exact free and periodic coefficients. Dantchev, Tonchev and Rudnick approximate these coefficients by Bessel functions when their variable $x_t = 2Nu$ is of order one, for periodic ends [4, Section 4.3.1 and Appendix D] and with relative error $O(1/N)$ for free ends [5, Section 3, Eq. (3.10)]. At our crossing, however, x_t grows like $2 \log N$, so we need control beyond that fixed scale.

The scaling variable of Antal, Droz and Rácz is our τ. Their law already suggests the crossing if we evaluate it at τ of order $\log N$ (although their limit keeps τ fixed). For free ends the endpoint atom is $e^{-\tau}/2$, while the density at zero magnetization is $\frac{1}{2}\tau e^{-\tau}(I_0(\tau) + I_1(\tau))$ by [2, Eq. (47)]. Note that neighboring magnetizations per spin are $2/N$ apart. Hence the middle bin and an endpoint should balance when the atom equals $2/N$ times this density, giving

$$\tau\big(I_0(\tau) + I_1(\tau)\big) = N/2.$$

This is the equation $F(x_B) = h$ for the Bessel root x_B of Theorem 3.3(ii), with $\tau = 2x_B$. Writing $z_B = 2x_B$, its large-argument expansion gives

$$z_B + \tfrac{1}{2} \log z_B = L + c_0 + \frac{1}{8z_B} + O(z_B^{-2}),$$

with the same displayed terms as the crossing equation of Proposition 3.8. We prove that the Bessel root approximates the actual crossing to relative error $O(L/N)$ in the odds (Theorem 3.3(iii)), give explicit finite-N bounds, and prove the bracket of Theorem 3.4. The known Bessel laws also suggest the $\log \log N$ term, the periodic shift by $\log 2$, and the Gaussian profile of Theorem 5.3. Our bounds turn these heuristics into theorems about the exact crossings.

Stepanyan et al. [6, Section III.B] study two crossings of the periodic chain as thermal transitions. Their first crossing compares the all-up configuration with magnetization $N - 2$ (one flipped spin), and is exactly at $J = \frac{1}{4} \log N$. Their second compares all up with zero magnetization, and they estimate its coupling. We give the second crossing with its constants for both boundary conditions. For free ends we also bound it at every N and find the crossing of every bin. The Markov binomial estimates of Čekanavičius and Mikalauskas assume ranges of transition probabilities that exclude

$p \to 1$ in this symmetric model (see [16, Theorems 1.1 to 1.3, pp. 220–221] and [17, Theorems 1.1 to 1.3, pp. 308–309]).

We prove relative Bessel bounds that are explicit in N, in u and in the bin (Theorems 3.3, 4.3 and 5.1), and that stay small when the mean number of domain walls grows like $\log N$. We bracket the free crossing at every N (Theorem 3.4). We expand the actual crossings at every growing distance a from the edge, uniformly over $b \geq a$ (Theorem 4.9), describe the crossing window about the exact root (Theorem 5.3), and locate both crossings in a fixed field (Theorem 5.9).

2 The exact law

The exact law and recurrence are classical, and so are the generating function [8, Eq. (3.1)] and the fixed-p normal limit [14, Theorem 6]. The count of Theorem 2.1 is Ising's count of domains [1], and it appears in [2, Eq. (44)] and [3, Eqs. (A15) to (A18)]. The law also appears in [15, Proposition 4.1 and Eq. (4.1)], and García-Pelayo [9] writes it in terms of ${}_2F_1$. We prove Theorem 2.1 because the later sections use its count of words by switches. By (1), a word with j switches has probability $\frac{1}{2}p^{N-1-j}q^j$.

Theorem 2.1 (bin probabilities)**.** *Let $N \geq 1$. Then $P_N(0) = P_N(N) = \frac{1}{2}p^{N-1}$, and for $1 \leq k \leq N-1$*

$$P_N(k) = \frac{1}{2}\sum_{j=1}^{N-1} W(N,k,j)\, p^{N-1-j}q^j,$$

where $W(N,k,j)$ counts the words with k letters $+$, $N-k$ letters $-$ and exactly j switches. Writing $a = k$ and $b = N-k$, for every integer $i \geq 1$

$$W(N,k,2i-1) = 2\binom{a-1}{i-1}\binom{b-1}{i-1}, \qquad W(N,k,2i) = \binom{a-1}{i}\binom{b-1}{i-1} + \binom{a-1}{i-1}\binom{b-1}{i}$$

with $\binom{n}{r} = 0$ for integers $n \geq 0$ and r outside $0 \leq r \leq n$.

Proof. By (1) it suffices to count the words in bin k with j switches. We count them by their run lengths. A word with j switches consists of $j+1$ maximal runs, alternately of letters $+$ and of letters $-$. If $k \in \{0, N\}$ there is one word and it has no switch. Let $a, b \geq 1$. If $j+1 = 2i$ there are i runs of each letter. The run lengths form a composition of a into i parts and a composition of b into i parts, and there are $\binom{a-1}{i-1}\binom{b-1}{i-1}$ such pairs. The two choices of the first letter give the factor 2. If $j+1 = 2i+1$ the word starts and ends with the same letter. Starting with $+$ gives $i+1$ runs of $+$ and i runs of $-$, so $\binom{a-1}{i}\binom{b-1}{i-1}$ words, and starting with $-$ gives the other summand. □

For instance $P_2 = (\frac{p}{2},\, q,\, \frac{p}{2})$, $P_3(1) = \frac{1}{2}(1-p^2)$ and $P_4(2) = q(1-p+p^2)$. Each interior bin needs at least one switch, while the endpoints need none. So the endpoints win when p is close to 1 (the walk rarely turns), and Section 3 asks how close.

Proposition 2.2 (recurrence and generating function)**.** *Let $P_N^+(k)$ and $P_N^-(k)$ be the probabilities of being in bin k after N steps with last letter $+$, respectively $-$, and set them to 0 outside $0 \leq k \leq N$. Then $P_1^+(1) = P_1^-(0) = \frac{1}{2}$, $P_1^+(0) = P_1^-(1) = 0$, and for $N \geq 2$*

$$P_N^+(k) = p\,P_{N-1}^+(k-1) + q\,P_{N-1}^-(k-1), \qquad P_N^-(k) = p\,P_{N-1}^-(k) + q\,P_{N-1}^+(k). \tag{3}$$

Consequently, in formal variables x and t,

$$\sum_{N\ge 0}\sum_{k=0}^{N} P_N(k)\,x^k t^N = \frac{1-(p-\frac12)(1+x)\,t}{1-p(1+x)\,t+(2p-1)\,x\,t^2}, \tag{4}$$

and for all $N \ge 2$ and all k, with $P_M(k) = 0$ outside $0 \le k \le M$,

$$P_N(k) = p\big(P_{N-1}(k) + P_{N-1}(k-1)\big) - (2p-1)\,P_{N-2}(k-1). \tag{5}$$

Proof. The recursion (3) follows by conditioning on w_{N-1}, since w_N equals w_{N-1} with probability p. Let $H^+ = \sum_{N,k} P_N^+(k)x^k t^N$ and define H^- alike. By (3), $H^+ = xt(\frac12 + pH^+ + qH^-)$ and $H^- = t(\frac12 + pH^- + qH^+)$. Solving these two linear equations and adding 1 for $N = 0$ gives (4), which is Gillis's generating function [8, Eq. (3.1)] after changing from position to bin number. Multiplying (4) by its denominator and comparing the coefficients of $x^k t^N$ with $N \ge 2$ gives (5). □

Remark 2.3 (integer numerators). For $p = \alpha/(\alpha+\beta)$ with integers $\alpha, \beta \ge 0$, not both 0, put $n_N(k) = 2(\alpha+\beta)^{N-1}P_N(k)$ for $N \ge 1$. By (1), $n_N(k)$ is the sum of $\alpha^{\mathrm{rep}(w)}\beta^{\mathrm{chg}(w)}$ over the words with k letters $+$ and $N-k$ letters $-$, which shows that it is an integer.

2.1 Variance and the normal limit

The bin K_N counts the visits of the chain $w_1, \dots, w_N$ to the letter $+$. This chain leaves each letter with probability q and starts from its stationary law $(\frac12, \frac12)$. So K_N has the Markov binomial law $\mathrm{Bin}(N, q, q, (\frac12, \frac12))$ of Dekking and Kong [15]. The transition matrix of the chain has second eigenvalue $1 - 2q = 2p - 1$.

Proposition 2.4 (mean and variance)**.** *Let $0 < p < 1$ and $\lambda = 2p - 1$. Then $\mathbb{E}K_N = N/2$ and*

$$\operatorname{Var} K_N = \frac14\Big(N\,\frac{1+\lambda}{1-\lambda} - \frac{2\lambda\,(1-\lambda^N)}{(1-\lambda)^2}\Big) = \frac{Np}{4(1-p)} - \frac{(2p-1)\big(1-(2p-1)^N\big)}{8(1-p)^2}.$$

Proof. This is [15, Eq. (2.3) and Proposition 2.1] with both of their switching probabilities equal to q and the stationary start, for which both excentricities vanish and both stationary weights are $\frac12$. The second form follows from $(1+\lambda)/(1-\lambda) = p/q$ and $1 - \lambda = 2q$. □

Theorem 2.5 (normal limit)**.** *Let $0 < p < 1$ be fixed. Then $(K_N - N/2)/\sqrt{N}$ converges in distribution, as $N \to \infty$, to the normal law with mean 0 and variance $p/(4(1-p))$.*

Proof. By (3), the vector $v_N(\theta)$ with entries $\mathbb{E}[e^{i\theta X_N}; \sigma_N = \pm 1]$ satisfies $v_{N+1} = T_\theta v_N$, where T_θ has rows $(pe^{i\theta}, qe^{i\theta})$ and $(qe^{-i\theta}, pe^{-i\theta})$. The eigenvalues of T_θ at $\theta = 0$ are 1 and $2p-1$, so for small $|\theta|$ they stay simple, with $\lambda_+(\theta) = p\cos\theta + \sqrt{q^2 - p^2\sin^2\theta} = 1 - \frac{p}{2q}\theta^2 + O(\theta^4)$ and λ_- near $2p-1$. Since $v_1(0)$ is an eigenvector for the eigenvalue 1, the component of $v_1(\theta)$ along the eigenvector of $\lambda_+(\theta)$ tends to 1 as $\theta \to 0$, the term with λ_-^N tends to 0 since $|\lambda_-| < 1$, and at $\theta = s/\sqrt{N}$ we get $\mathbb{E}e^{i\theta X_N} \to \exp(-\frac{p}{2q}s^2)$. Lévy's continuity theorem [18, Theorem 26.3] gives the normal limit of $X_N/\sqrt{N}$ with variance p/q, and $K_N - N/2 = X_N/2$. □

Compared with the simple random walk, persistence multiplies the width of the bell by $\sqrt{p/q}$, so it widens the bell for $p > \frac12$ and narrows it for $p < \frac12$.

3 The central crossing at every N

In this section the chain has free ends. Let $N \geq 2$, let $m_N = \lfloor N/2 \rfloor$ and let $p > 0$. By Theorem 2.1 the ratio of the middle bin to an endpoint is

$$\frac{P_N(m_N)}{P_N(0)} = S_N(u) := \sum_{j=1}^{N-1} W(N, m_N, j)\, u^j, \tag{6}$$

a polynomial in u with nonnegative integer coefficients and $W(N, m_N, 1) = 2$. The middle and the endpoints are level exactly when $S_N(u) = 1$.

We first explain heuristically why this happens when τ is near $\log N$. Let $a = m_N$ and $b = N - m_N$, both about $N/2$. A word in the middle bin with $2r+1$ switches has $r+1$ runs of each letter, and there are $2\binom{a-1}{r}\binom{b-1}{r} \approx 2(N/2)^{2r}/(r!)^2$ of them when r is small compared with $\sqrt{N}$. So these words contribute about $2u\sum_r (Nu/2)^{2r}/(r!)^2 = 2uI_0(Nu)$ to $S_N(u)$, and in the same way the words with an even number of switches contribute about $2uI_1(Nu)$. For large $w = Nu$ we have $I_0(w) + I_1(w) \approx 2e^w/\sqrt{2\pi w}$, so $S_N(u) \approx 4we^w/(N\sqrt{2\pi w})$. Setting this equal to 1 gives $w + \frac{1}{2}\log w \approx L + \frac{1}{2}\log(\pi/8)$. So at the crossing w, and with it τ, is close to $L - \frac{1}{2}\log L + c_0$ with $c_0 = \frac{1}{2}\log(\pi/8)$.

Theorem 3.1. *(a) For every $N \geq 2$ there is exactly one $p = p_N \in [0,1]$ with $P_N(m_N) = P_N(0)$, and $p_N \geq \frac{2}{3}$. The middle is more likely than the endpoints for $p < p_N$ and less likely for $p > p_N$.*

(b) The first values are $p_2 = \frac{2}{3}$, $p_3 = 1/\sqrt{2}$ and the real root p_4 of $3p^3 - 4p^2 + 4p - 2$, and $p_N < p_{N+1}$ for all $N \geq 2$.

(c) As $N \to \infty$ we have $p_N \to 1$, and more precisely $1 - p_N \sim (\log N)/N$.

Proof. (a) At $p = 0$ the endpoints have probability 0 and the middle does not. For $p > 0$ the equation is $S_N(u) = 1$ by (6). The coefficients of S_N are nonnegative and the linear one is 2, so S_N is strictly increasing on $[0, \infty)$ with $S_N(0) = 0$ and $S_N(u) \geq 2u$. By the intermediate value theorem there is exactly one root $u_N \in (0, \frac{1}{2}]$. The map $u \mapsto p = 1/(1+u)$ is a decreasing bijection from $[0, \infty)$ to $(0, 1]$. Hence $p_N = 1/(1 + u_N) \geq \frac{2}{3}$, and the middle wins exactly when $u > u_N$.

(b) By Theorem 2.1, $S_2 = 2u$, $S_3 = 2u + u^2$ and $S_4 = 2u + 2u^2 + 2u^3$, which give the three values. For p_4, substitute $u = (1-p)/p$ in $S_4(u) = 1$. For the monotonicity we compare coefficients. Going from N to $N+1$, one entry of the middle pair $(a, b) = (\lfloor N/2 \rfloor, \lceil N/2 \rceil)$ grows by one and the other is unchanged. The formulas of Theorem 2.1 are symmetric in (a, b), and every $\binom{a-1}{\cdot}$ is nondecreasing in a. So $W(N+1, m_{N+1}, j) \geq W(N, m_N, j)$ for every j. The coefficient of u^2 is $a + b - 2 = N - 2$, which increases strictly. Hence $S_{N+1}(u) > S_N(u)$ for $u > 0$, so $u_{N+1} < u_N$.

(c) We prove (c) after Theorem 3.3, from part (i) of that theorem. □

Figure 1 shows the law of a chain of nine spins below, at and above its crossing.

3.1 A Bessel bound at every N

The heuristic above replaced the number $\binom{a-1}{r}$ of ways to split a letters into $r+1$ runs by $a^r/r!$. After this change the run count is exactly the power series of I_0 and I_1. The next lemma bounds the error this makes.

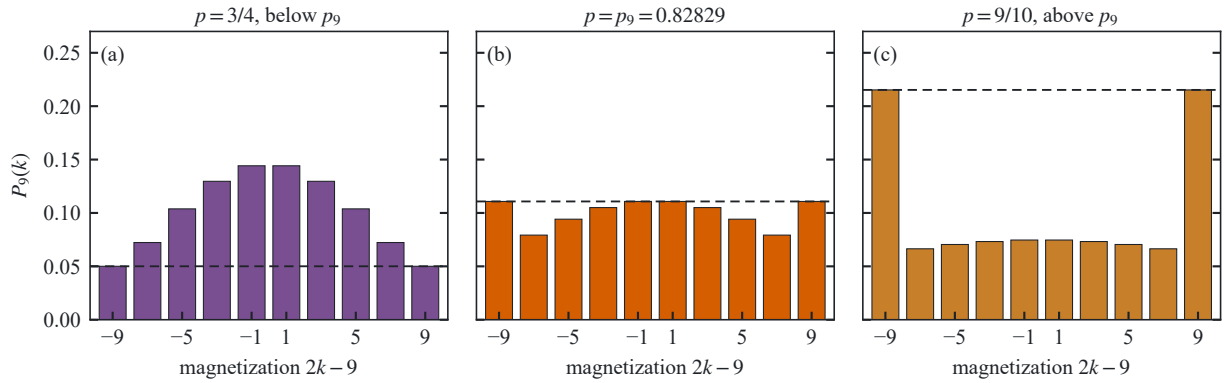


Figure 1: The bin probabilities $P_9(k)$ of a chain of nine spins against the magnetization $2k-9$, below (a), at (b) and above (c) the crossing $p_9 = 0.82829\ldots$, on a common vertical scale. The dashed line is the endpoint height. At $p = p_9$ the two central bins, with magnetization -1 and 1, are exactly as likely as each endpoint.

Lemma 3.2 (clipped products)**.** *(a) If $t_1, \ldots, t_r \geq 0$ and $(v)_+ = \max(v, 0)$, then $0 \leq 1 - \prod_{k=1}^{r}(1 - t_k)_+ \leq \sum_{k=1}^{r} t_k$. (b) For integers $m \geq 1$ and $r \geq 0$, $U_r(m) := \frac{r!}{m^r}\binom{m-1}{r} = \prod_{k=1}^{r}\bigl(1 - \frac{k}{m}\bigr)_+$, so $0 \leq 1 - U_r(m) \leq \frac{r(r+1)}{2m}$. (c) For $x \geq 0$,*

$$\sum_{r\geq 0} \frac{r(r+1)x^{2r}}{(r!)^2} = x^2 I_0(2x) + x I_1(2x),$$

$$\sum_{r\geq 0} \frac{r(r+1)x^{2r+1}}{r!(r+1)!} = x^2 I_1(2x), \qquad \sum_{r\geq 0} \frac{(r+1)x^{2r+1}}{r!(r+1)!} = x I_0(2x).$$

Proof. (a) Every factor lies in $[0, 1]$, so the product is at most 1. If some $t_k \geq 1$ the product is 0 and the sum is at least 1. Otherwise $\prod_k (1 - t_k) \geq 1 - \sum_k t_k$ by induction on r. (b) Note that $\frac{r!}{m^r}\binom{m-1}{r} = \prod_{k=1}^{r} \frac{m-k}{m}$. For $r < m$ every factor is positive, and for $r \geq m$ both sides vanish, the right side through the factor $k = m$. The bound is (a) with $t_k = k/m$. (c) Use the series of $I_0(2x)$ and $I_1(2x)$ from Section 1.2. Write $r(r+1) = r^2 + r$ in the first sum, cancel the factors r and $r+1$ against the factorials, and shift the index. □

Theorem 3.3 (uniform Bessel approximation)**.** *Put $h = N/2$ and $F(x) = 2x\bigl(I_0(2x) + I_1(2x)\bigr)$, and recall that $u_N = (1 - p_N)/p_N$.*

(i) For every $N \geq 2$ and $x > 0$,

$$0 \leq 1 - \frac{hS_N(x/h)}{F(x)} \leq \frac{x(x+1)}{h}. \tag{7}$$

(ii) Let $x_N = hu_N$, let x_B be the unique positive root of $F(x_B) = h$, and put $\delta_N = x_N(x_N+1)/h$. Then $x_B \leq x_N$, and $x_N - x_B \leq -\frac{1}{2}\log(1 - \delta_N)$ whenever $\delta_N < 1$.

(iii) With $u_B = x_B/h$ we have $0 \leq u_N - u_B = O(L^2/N^2)$ as $N \to \infty$.

Section 3.2 uses part (i) to locate p_N at every N.

Proof. (i) *Step 1. The clipped factors.* Let $a = \lfloor N/2 \rfloor$ and $b = \lceil N/2 \rceil$. As in Lemma 3.2(b), $A_r = \frac{r!}{h^r}\binom{a-1}{r} = \prod_{k=1}^{r}(1 - \frac{h-a+k}{h})_+$, and B_r is the same with b in place of a. Since $a + b = 2h$ and

$h-a \in \{0, \frac{1}{2}\}$, the $2r$ numbers $(h-a+k)/h$ and $(h-b+k)/h$ are nonnegative with sum $r(r+1)/h$, so Lemma 3.2(a) gives $0 \le 1 - A_r B_r \le r(r+1)/h$. In the same way the deficits of $A_{r+1}B_r$ and $A_r B_{r+1}$ are at most $((r+1)^2 + h - a)/h$ and $((r+1)^2 + h - b)/h$. Averaging cancels the parity offsets (the shifts $h-a$ and $h-b$ add to 0), so $0 \le 1 - \frac{1}{2}(A_{r+1}B_r + A_r B_{r+1}) \le (r+1)^2/h$.

Step 2. Summing. With $u = x/h$ the run-count formula of Theorem 2.1 becomes

$$S_N(u) = 2u\Big[\sum_{r\ge 0} \frac{x^{2r}}{(r!)^2} A_r B_r + \sum_{r\ge 0} \frac{x^{2r+1}}{r!(r+1)!} \, \frac{A_{r+1}B_r + A_r B_{r+1}}{2}\Big].$$

Without the factors A_r, B_r the bracket is $I_0(2x)+I_1(2x)$ and the right side is $F(x)/h$. By Step 1 and Lemma 3.2(c), $F(x)/h - S_N(u)$ lies between 0 and $2u/h$ times $\sum_r r(r+1)\frac{x^{2r}}{(r!)^2} + \sum_r (r+1)^2 \frac{x^{2r+1}}{r!(r+1)!} = x(x+1)\big(I_0(2x) + I_1(2x)\big)$. Dividing by $F(x)/h = 2u(I_0(2x) + I_1(2x))$ proves (i). This covers every r, including those beyond the degree of the polynomial, where the clipped products vanish and Lemma 3.2(a) still applies.

(ii) Differentiating the series term by term gives $F' = 2F + 2I_0(2x)$. So F increases from 0 to ∞, x_B is unique, and $(\log F)' \ge 2$. At $x = x_N$ we have $hS_N(u_N) = h$, so (i) gives $h \le F(x_N)$, hence $x_B \le x_N$, and $F(x_N) \le h/(1-\delta_N)$ when $\delta_N < 1$. Integrating $(\log F)' \ge 2$ from x_B to x_N gives $2(x_N - x_B) \le \log(F(x_N)/h) \le -\log(1-\delta_N)$.

Proof of Theorem 3.1(c). We use only part (i). Note that $x^{2r}/(r!)^2 \le (2x)^{2r}/(2r)!$ and $x^{2r+1}/(r!(r+1)!) \le (2x)^{2r+1}/(2r+1)!$ for $x \ge 0$, since $\binom{2r}{r} \le 4^r$ and $\binom{2r+1}{r} \le 2^{2r+1}$. So $I_0(2x) + I_1(2x) \le e^{2x}$ and $F(x) \le 2xe^{2x}$. Fix $\theta \in (0,1)$ and put $x_\pm = (1 \pm \theta)L/2$. Then $F(x_-) \le (1-\theta)LN^{1-\theta} < h$ for large N, so $hS_N(x_-/h) < h$ by (i), and $x_- < x_N$ since S_N increases. On the other side $F(x_+) \ge 2x_+ I_0(2x_+)$, which is at least a constant times $\sqrt{L}\, N^{1+\theta}$ for large N by the large-argument form of I_0, while $x_+(x_+ + 1)/h \to 0$. So $hS_N(x_+/h) \ge F(x_+)(1 - x_+(x_+ + 1)/h) > h$ for large N, and $x_N < x_+$. Hence $Nu_N = 2x_N \sim L$, so $u_N \to 0$ and $z_N = Nu_N/(1+u_N) \sim L$, which is (c).

(iii) By Theorem 3.1(c), $x_N \sim L/2$. So $\delta_N = O(L^2/N)$, and (ii) gives $x_N - x_B = O(L^2/N)$. Divide by h. □

3.2 The crossing at every N

We measure the crossing by $z_N = N(1-p_N) = Nu_N/(1+u_N)$, the value of τ at $p = p_N$. Let $\psi(z) = z + \frac{1}{2}\log z$ and $c_0 = \frac{1}{2}\log(\pi/8) = -0.46735\ldots$, and let ζ_N be the positive root of

$$\psi(\zeta_N) = L + c_0, \quad \text{that is,} \quad 8\zeta_N e^{2\zeta_N} = \pi N^2 \quad \text{or} \quad \zeta_N = \tfrac{1}{2} W_0(\pi N^2/4), \tag{8}$$

with W_0 the principal branch of the Lambert function. This is the heuristic equation from the start of the section with w replaced by ζ. Since $\zeta e^{2\zeta}$ increases, ζ_N increases with N. Also $\zeta_N \le L$, since $\psi(L) \ge L + c_0$ when $L \ge \pi/8$. Finally let $\varepsilon_N = \zeta_N(\zeta_N+1)/N + 1/(16\zeta_N^2)$.

Theorem 3.4 (the crossing at every N)**.** *For every $N \ge 2$,*

$$\zeta_N + \frac{1}{8\zeta_N} - \varepsilon_N < z_N < \zeta_N + \frac{1}{8\zeta_N}.$$

The width ε_N tends to 0, since $\zeta_N \le L$ and $\zeta_N \to \infty$. It is 0.440, 0.163, 0.0387 and 0.00778 at $N = 10, 10^2, 10^3, 10^4$ (Table 1).

The two terms ζ_N and $1/(8\zeta_N)$ come from the following heuristic. Let $\Phi(w) = F(w/2) = w\big(I_0(w) + I_1(w)\big)$ and $\mathcal{E}(z) = \sqrt{\pi z/2}\, e^{-z}\big(I_0(z) + I_1(z)\big)$, which tends to 1 as $z \to \infty$. Put $w_* = Nu_N$

and $\Lambda_N = 2\Phi(w_*)/N$. At the crossing, (7) and (9) below give $\psi(w_*) + \log \mathcal{E}(w_*) = L + c_0 + \log \Lambda_N$ with $0 \le \log \Lambda_N \le -\log(1 - w_*(w_*+2)/(2N))$ whenever $w_*(w_*+2) < 2N$. With $\mathcal{E} = \Lambda_N = 1$ this would say $w_* = \zeta_N$. Since $\log \mathcal{E}(w) = -1/(8w) + O(w^{-2})$ and $z_N = w_* + O(w_*^2/N)$, it suggests $z_N = \zeta_N + 1/(8\zeta_N) + O(\zeta_N^{-2} + \zeta_N^2/N)$. The theorem makes this error explicit and one-sided. Its proof uses two lemmas. Lemma 3.5 turns a comparison of z_N with a test value into a sign condition, and Lemma 3.6 bounds $\mathcal{E}$ with explicit constants.

Lemma 3.5 (root bracket)**.** *Let $N \ge 2$ and $w > 0$. Then*

$$\log \Phi(w) - \log \tfrac{N}{2} = \psi(w) - \psi(\zeta_N) + \log \mathcal{E}(w). \tag{9}$$

(a) If $\Phi(w) \le N/2$, then $Nw/(N+w) \le z_N$, strictly if $\Phi(w) < N/2$. (b) If $\Phi(w)\big(1 - \frac{w(w+2)}{2N}\big) > N/2$, then $z_N < Nw/(N+w)$.

Proof. Note that $\log \Phi(w) = \psi(w) + \frac{1}{2}\log\frac{2}{\pi} + \log \mathcal{E}(w)$ and $\log \frac{N}{2} - \psi(\zeta_N) = -\log 2 - c_0 = \frac{1}{2}\log\frac{2}{\pi}$, which gives (9). Now apply (7) with $x = w/2$. Then $F(x) = \Phi(w)$ and $x(x+1)/h = w(w+2)/(2N)$, so

$$\Phi(w)\Big(1 - \frac{w(w+2)}{2N}\Big) \le \frac{N}{2}\, S_N\Big(\frac{w}{N}\Big) \le \Phi(w).$$

(a) The hypothesis gives $S_N(w/N) \le 1 = S_N(u_N)$, strictly if $\Phi(w) < N/2$. Since S_N is strictly increasing, $w/N \le u_N$, strictly in that case. The map $v \mapsto Nv/(1+v)$ increases and sends w/N to $Nw/(N+w)$ and u_N to z_N. (b) Now $S_N(w/N) > 1$, so $w/N > u_N$. □

Lemma 3.6 (Bessel bounds)**.** *For every $z > 0$,*

$$1 - \frac{1}{8z} - \frac{3}{128z^2} - \frac{45}{512z^3} \le \mathcal{E}(z) \le 1 - \frac{1}{8z} - \frac{3}{128z^2} + \frac{15}{512z^3},$$

and $\mathcal{E}(z) \le 1$. In particular $\mathcal{E}(z) \le 1 - \frac{1}{8z}$ for $z \ge \frac{5}{4}$.

The lemma is the large-argument expansion of $I_0 + I_1$ with explicit errors. We prove it in Appendix A by bounding $\sqrt{1-y}$ by cubics in an integral form of $\mathcal{E}$.

Proof of Theorem 3.4. Step 1. The split at $N = 175$. By (8), $N = \sqrt{8\zeta_N/\pi}\, e^{\zeta_N}$, and the right side increases in ζ_N. Since $\sqrt{32/\pi}\, e^4 = 174.25\ldots$, we have $\zeta_N \ge 4$ exactly when $N \ge 175$. Write $\zeta = \zeta_N$ and $Y = \zeta + \frac{1}{8\zeta}$.

Step 2. The upper bound for $N \ge 175$. We apply Lemma 3.5(b) at $\bar{w} = NY/(N-Y)$, for which $N\bar{w}/(N+\bar{w}) = Y$. By (9) it suffices that $\mathcal{G} = \psi(\bar{w}) - \psi(\zeta) + \log \mathcal{E}(\bar{w}) + \log(1 - \bar{w}(\bar{w}+2)/(2N)) > 0$. Expanding each term with Lemma 3.6 gives $\mathcal{G} \ge \frac{1}{32\zeta^2} + \frac{\zeta(\zeta-1)}{2N} + O(\zeta^{-3} + N^{-1} + \zeta^4 N^{-2})$, and Appendix A makes every term explicit for $\zeta \ge 4$.

Step 3. The lower bound for $N \ge 175$. We apply Lemma 3.5(a) at $\bar{w}' = NY'/(N - Y')$ with $Y' = Y - \varepsilon_N$. It suffices that $\mathcal{G}' = \psi(\bar{w}') - \psi(\zeta) + \log \mathcal{E}(\bar{w}') < 0$, and we get $\mathcal{G}' \le -\frac{3}{128\zeta^2} - \frac{\zeta}{N} + O(\zeta^{-3} + N^{-1} + \zeta^4 N^{-2})$, again with explicit constants for every $\zeta \ge 4$ in Appendix A. The term $-\zeta/N$ is why ε_N contains $\zeta(\zeta+1)/N$.

Step 4. Short chains, by certificates. Here $2 \le N \le 174$. Let $d = 10^9$, let $c = \lfloor du_N \rfloor$ and let $c' = c+1$. At $N = 2$, where $du_2 = 5 \cdot 10^8$ is an integer, we take $c = du_2 - 1$ and $c' = c+2$. The sign of $S_N(c/d) - 1$ is the sign of $n_N(m_N) - d^{N-1}$, where $n_N(m_N) = d^{N-1} S_N(c/d)$ is the integer of Remark 2.3 at $(\alpha, \beta) = (d, c)$ (Appendix A.3). We computed these integers at c and c' by two exact methods, the run-count sum of Theorem 2.1 and the recursion (3) with integer weights d, c in place of p, q. They agreed for every $N \le 174$ and gave $c/d < u_N < c'/d$, so $z(c/d) < z_N < z(c'/d)$ with

$z(u) = Nu/(1+u)$. We then checked $\zeta_N + \frac{1}{8\zeta_N} - \varepsilon_N < z(c/d)$ and $z(c'/d) < \zeta_N + \frac{1}{8\zeta_N}$ with outward interval arithmetic at 80 decimal digits. To enclose π, we used $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ and 80 terms of each alternating series, with the first omitted term bounding its remainder. Each arithmetic operation rounds outward. For a logarithm, we take the neighboring decimal on each side of the correctly rounded endpoint value.

Newton's method only proposes a value of ζ_N. We certify the bracket of half-width 10^{-45} around it by enclosing $\psi(\zeta) - L - c_0$ at both ends and checking opposite signs. Evaluating both comparisons on this whole bracket gives margins greater than 0.03319 (upper) and 0.08559 (lower), with the smallest at $N = 174$. The script `verify_certified_bounds.py` replays these checks from the stored rational endpoints, including all rounding and remainder bounds. Lean checks the short-chain comparisons independently with rational bounds on ζ_N and π. We also ran the certificates up to $N = 1000$ as a sanity check (both methods up to 400, then only the run-count sum), and they all pass. □

N	z_N	ζ_N	$z_N - \zeta_N - \frac{1}{8\zeta_N}$	$-\varepsilon_N$	R_N
2	0.666666666667	0.5368290974	−0.1030112	−0.62938	+0.050643
10	1.615958871376	1.6001732978	−0.0623310	−0.44048	−0.056660
100	3.503367368092	3.5100055489	−0.0422507	−0.16337	−0.043265
174	3.996779704730	3.9987131703	−0.0331935	−0.11878	−0.033450
175	4.001967989170	4.0038072816	−0.0330596	−0.11838	−0.033304
1000	5.590351166975	5.5807388611	−0.0127862	−0.03873	−0.011887
10^4	7.734265514999	7.7210118331	−0.0029359	−0.00778	−0.002051

Table 1: The crossing against Theorem 3.4, and the remainder R_N of (12). We computed z_N by run-count bisection in 50-digit decimal arithmetic and by a floating-point dynamic program, which agree to the digits shown.

The bracket is in terms of ζ_N (a value of the Lambert function). The next corollary states it for p_N and expands it in $L = \log N$.

Corollary 3.7. *Put $\nu_N = L - \frac{1}{2}\log L + c_0$, $Y_N = \zeta_N + \frac{1}{8\zeta_N}$ and $Y'_N = Y_N - \varepsilon_N$.*

(a) For every $N \geq 2$ we have $0 < Y'_N < Y_N < N$, $1 - Y_N/N < p_N < 1 - Y'_N/N$ and, for the coupling $J_N^{\mathrm{F}} = -\frac{1}{2}\log u_N$ at the crossing,

$$\frac{1}{2}\log\Big(\frac{N}{Y_N} - 1\Big) < J_N^{\mathrm{F}} < \frac{1}{2}\log\Big(\frac{N}{Y'_N} - 1\Big).$$

(b) For every $N \geq 2$, $\nu_N \leq \zeta_N \leq \nu_N + \frac{\log L - 2c_0}{4\nu_N}$ and

$$-\varepsilon_N < z_N - \nu_N < \frac{\log L - 2c_0 + \frac{1}{2}}{4\nu_N}. \tag{10}$$

In particular $z_N = L - \frac{1}{2}\log L + c_0 + O(\log L/L)$ as $N \to \infty$.

(c) As $N \to \infty$,

$$z_N = \nu_N + \frac{\frac{1}{4}\log L - \frac{1}{2}c_0 + \frac{1}{8}}{L} + O\Big(\frac{(\log L)^2}{L^2}\Big). \tag{11}$$

(d) (Computer-assisted.) $z_N > \zeta_N$ for every $N \geq 2774$.

Proof. (a) Since $\frac{1}{2} < \zeta_2 \le \zeta_N$, we have $\frac{1}{16\zeta_N^2} < \frac{1}{8\zeta_N}$, and since $\zeta_N \le L < N-1$, we have $\zeta_N(\zeta_N+1)/N < \zeta_N$. So $\varepsilon_N < Y_N$, that is, $Y_N' > 0$. Also $Y_N < L + \frac{1}{4} < N$. The bounds are Theorem 3.4 with $p_N = 1 - z_N/N$ and, by (2), $J_N^{\mathrm{F}} = \frac{1}{2}\log(N/z_N - 1)$, which decreases in z_N.

(b) By (8), $\zeta_N = L + c_0 - \frac{1}{2}\log\zeta_N$. Since $\zeta_N \le L$, this gives $\zeta_N \ge \nu_N$, and $\nu_N > 0$ since it increases in N and $\nu_2 > 0.40$. Also $\log y \le y - 1$ gives $\zeta_N - \nu_N = \frac{1}{2}\log(L/\zeta_N) \le (L - \zeta_N)/(2\zeta_N)$, and $L - \zeta_N = \frac{1}{2}\log\zeta_N - c_0 \le \frac{1}{2}\log L - c_0$. Hence $\zeta_N - \nu_N \le (\log L - 2c_0)/(4\nu_N)$. For z_N, add Theorem 3.4 and $0 < \frac{1}{8\zeta_N} \le \frac{1}{8\nu_N}$. The last claim follows since $\varepsilon_N = O(L^2/N + L^{-2})$.

(c) Put $\mu = \zeta_N - \nu_N = \frac{1}{2}\log(L/\zeta_N)$. By (b), $L - \zeta_N = \frac{1}{2}\log L - c_0 + O(\log L/L)$, so $\mu = \frac{L-\zeta_N}{2L} + O\big(\frac{(\log L)^2}{L^2}\big) = \frac{\frac{1}{4}\log L - \frac{1}{2}c_0}{L} + O\big(\frac{(\log L)^2}{L^2}\big)$. Also $\frac{1}{8\zeta_N} = \frac{1}{8L} + O(\frac{\log L}{L^2})$ and $\varepsilon_N = O(L^2/N + L^{-2})$, and Theorem 3.4 gives (c).

(d) By Theorem 3.4, $z_N - \zeta_N > \frac{1}{8\zeta_N} - \varepsilon_N$, which is positive when $f(\zeta_N) > 1$, where $f(\zeta) = n(\zeta)\big(\frac{1}{8\zeta} - \frac{1}{16\zeta^2}\big)/(\zeta(\zeta+1))$ and $n(\zeta) = \sqrt{8\zeta/\pi}\,e^{\zeta}$, so that $n(\zeta_N) = N$. The logarithmic derivative of f is $1 + \frac{1}{2\zeta} - \frac{2}{\zeta} + \frac{1}{2\zeta^2-\zeta} - \frac{1}{\zeta+1} > 1 - \frac{3}{2\zeta} - \frac{1}{\zeta+1} > 0$ for $\zeta \ge 3$. Since ζ_N increases with N, once $f(\zeta_N) > 1$ it stays above 1. This step is computer-assisted. With the outward intervals of Step 4, we obtain $f(\zeta_{2774}) - 1 > 2.0 \times 10^{-4}$ on the whole bracket. Here we evaluate $f(\zeta_N)$ by replacing $n(\zeta_N)$ with the exact integer N. Since $\zeta_{2774} > 3$, this proves (d). The same interval check gives $f(\zeta_{2773}) < 1$. The conclusion for every $N \ge 2774$ is also proved in Lean. That proof uses rational Taylor bounds for ζ_{2774} and an exponential comparison to extend the bound to larger N. □

The heuristic before Lemma 3.5 suggests that $\psi(z_N) \approx L + c_0 + 1/(8z_N)$. The next proposition proves this at every N with an explicit remainder.

Proposition 3.8 (the crossing equation)**.** *For every $N \ge 2$,*

$$z_N + \tfrac{1}{2}\log z_N = L + c_0 + \frac{1}{8z_N} + R_N, \qquad -\frac{z_N^2 + z_N}{N} \le R_N \le \frac{1}{16z_N^2}. \tag{12}$$

We also have $L/2 \le z_N \le L$, so $|R_N| \le \frac{1}{4L^2} + \frac{L^2+L}{N}$ for every $N \ge 2$.

We prove Proposition 3.8 in Appendix A. The proof writes R_N exactly, using (7) and (9) at $w = w_*$ (formula (41)). For $N \ge 175$, Lemma 3.6 puts the Bessel part of this formula between 0 and $0.0559/z_N^2$, and the other terms lie between $-(z_N^2 + z_N)/N$ and 0. For $N \le 174$ the proof uses the same certificates as Theorem 3.4.

Remark 3.9 (the sign of $z_N - \zeta_N$, computer-assisted). We have $z_N > \zeta_N$ for $N \le 15$, $z_N < \zeta_N$ for $16 \le N \le 218$, and $z_N > \zeta_N$ for every $N \ge 219$. For $N \le 2773$ this follows from the sign of $S_N - 1$ in exact integers at rationals that separate u_N from $\zeta_N/(N - \zeta_N)$, the odds at which $z(u) = \zeta_N$, which the scripts list and replay with outward intervals (Section 7). The closest case is $N = 218$, where ζ_N is more than 1.65×10^{-5} from the certified bracket of z_N. For $N \ge 2774$ the sign is Corollary 3.7(d).

The ratios u_B/u_N of the Bessel root of Theorem 3.3 to the true odds are 0.9761, 0.9864, 0.9924 at $N = 100, 200, 400$ and 0.99579, 0.99769, 0.99874 at $N = 800, 1600, 3200$. So $1 - u_B/u_N$ is close to $L/(2N)$, which is 0.0230 at $N = 100$ and 0.00126 at $N = 3200$. This is in line with Theorem 3.3(iii), which gives a relative error $O(L/N)$ since $u_N \sim L/N$. Figure 2 compares the exact crossing with the expansions of Corollary 3.7 up to $N = 10^4$.

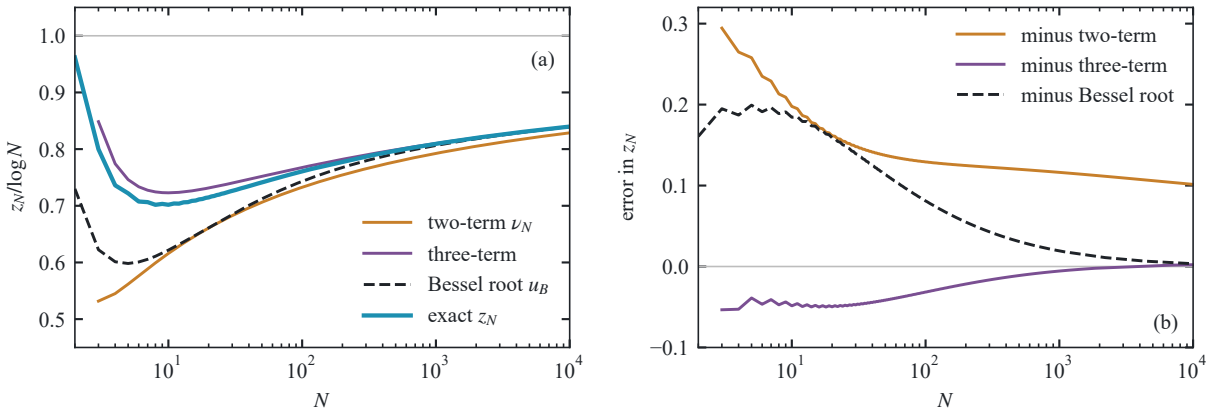


Figure 2: (a) The scaled crossing $z_N/\log N$ from the exact run polynomial (thick line), and the same quantity from the Bessel root u_B of Theorem 3.3 (dashed), from the two-term expansion ν_N of (10) and from the three-term expansion $\nu_N + (\frac{1}{4}\log L - \frac{1}{2}c_0 + \frac{1}{8})/L$ of (11). The ratio tends to 1 very slowly. (b) The exact z_N minus each approximation. The small oscillations for N below about 30 come from the parity of N.

4 The crossing of every bin

For each bin a of the free chain we find the value $p_{a,N-a}$ of p at which bin a is exactly as likely as an endpoint (Section 5.1 treats periodic ends). The switching probability $1 - p_{a,N-a}$ at the crossing, shown in Figure 3(a) for three values of N, decreases as the bin moves in from the edge, from order $N^{-1/2}$ next to the edge to order $(\log N)/N$ in the middle.

4.1 Ordering the crossings

Let $a, b \geq 1$ be integers, $N = a + b$ and $p > 0$, and recall that $S_{a,b}(u) = P_N(a)/P_N(0)$ with $u = (1-p)/p$. Theorem 2.1 and $\binom{m-1}{r+1} = \frac{m-1-r}{r+1}\binom{m-1}{r}$ for $m = a, b$ give

$$S_{a,b}(u) = \sum_{r\geq 0} \binom{a-1}{r}\binom{b-1}{r}\Big(2u^{2r+1} + \frac{N-2-2r}{r+1}u^{2r+2}\Big). \tag{13}$$

Its coefficients are nonnegative, as $N - 2 - 2r \geq |b-a|$ for $r < \min(a,b)$, and the linear one is 2. So there is a unique $u_{a,b} > 0$ with $S_{a,b}(u_{a,b}) = 1$, and $p_{a,b} = 1/(1+u_{a,b})$. As $S_{a,b}(u) \geq 2u$, we have $u_{a,b} \leq \frac{1}{2}$ and $p_{a,b} \in [\frac{2}{3}, 1)$, with $p_{a,b} = \frac{2}{3}$ only for $a = b = 1$.

Proposition 4.1 (ordering the crossings)**.** *If $a < b - 1$, then*

$$u_{a+1,b-1} < u_{a,b}, \qquad p_{a+1,b-1} > p_{a,b}.$$

Also $p_{a,b+1} > p_{a,b}$ for all positive a, b. In addition, $S_{a,b+1} \geq S_{a,b}$ coefficientwise, and if $a < b - 1$ then $S_{a+1,b-1} \geq S_{a,b}$ coefficientwise, in each case with at least one strict coefficient.

So a bin closer to the middle, or in a longer row, needs a larger p to fall to the endpoint height.

Proof. For fixed N and $r \leq a-1$, $\binom{a-1}{r}\binom{b-1}{r} = (r!)^{-2}\prod_{j=0}^{r-1}(a-1-j)(b-1-j)$, a product of positive factors. Each factor grows when (a,b) becomes $(a+1,b-1)$, by $(a-j)(b-2-j)-(a-1-j)(b-1-j) = b-a-1 > 0$. The multiplier $(N-2-2r)/(r+1)$ of u^{2r+2} is unchanged, and it is nonnegative where

either product is nonzero, since then $r \le a$ and $N-2-2r \ge b-a-2 \ge 0$. New summands are nonnegative, and the cubic coefficient grows strictly, from $2(a-1)(b-1)$ to $2a(b-2)$. Increasing b with a fixed decreases no coefficient of (13) and increases the quadratic coefficient $N-2$ strictly. A polynomial with nonnegative coefficients that dominates another coefficientwise, strictly somewhere, is strictly larger at every $u > 0$, so its solution of $S = 1$ is smaller. □

The ordering of the bin heights in Proposition 4.2 also follows from the interior log-concavity of Dekking and Kong [15, Proposition 4.3], with the symmetry of the row, and they remark after that proposition that a sharper proof gives strict log-concavity. We use the coefficients because they give the strict inequalities directly and also compare different rows. The formula for p_N^{adj} in Proposition 4.2 is already implicit in their work. For $N \ge 4$ and $0 < p < 1$ we have $P_N(1) < P_N(2)$, so by [15, Theorem 4.1] the row is trimodal exactly when $P_N(0) > P_N(1)$. Their Example 4.2, with both switching probabilities equal to q and the stationary start, gives $P_N(1)/P_N(0) = 2u + (N-2)u^2$, which is $S_{1,N-1}(u)$. Zheng, Lin and Zhang [19, Eqs. (7) to (9) of the arXiv version] derive the counts of Theorem 2.1 by Ising's method and the coefficientwise comparison of Proposition 4.1, and conclude that the Landau free energy per spin is smallest at zero magnetization in the thermodynamic limit. For $p > p_N^{\mathrm{adj}}$ the row is not monotone near its edges, since the endpoints are more likely than the adjacent bins. So at p_N the bin probabilities have their maxima at the edges and in the middle. The next proposition proves this, with the comparison $p_N > p_N^{\mathrm{adj}}$ for $N \ge 4$.

Proposition 4.2 (adjacent bins and global extrema)**.** *For $N \ge 2$ and $0 < p < 1$, an adjacent bin has the same probability as an endpoint precisely at*

$$p_N^{\mathrm{adj}} = \frac{1+\sqrt{N-1}}{2+\sqrt{N-1}}.$$

For $0 < p < 1$, the endpoints are the least likely bins if $p < p_N^{\mathrm{adj}}$, bins 1 and $N-1$ are if $p > p_N^{\mathrm{adj}}$, and they tie at equality. For $N \ge 4$, the central crossing satisfies $p_N > p_N^{\mathrm{adj}}$. At $p = p_N$, the middle bin and the endpoints attain the global maximum and bins 1 and $N-1$ attain the global minimum. For $N = 2$ and $N = 3$ we have $p_N = p_N^{\mathrm{adj}}$, and every bin ties at the crossing.

So for $N \ge 4$ the row at the central crossing has the shape of a W. It is high at the two edges and in the middle, and lowest next to the edges.

Proof. By (13), $S_{1,N-1}(u) = 2u + (N-2)u^2$, with positive root $u^{\mathrm{adj}} = 1/(1+\sqrt{N-1})$. This gives p_N^{adj} and the sign change. For $0 < p < 1$ we have $P_N(k) = P_N(0)S_{k,N-k}(u)$ with $P_N(0) > 0$, so if $1 \le k < N-k-1$, then Proposition 4.1 gives $P_N(k) < P_N(k+1)$. By symmetry the middle is the largest interior bin and bins $1, N-1$ are the smallest, which gives the claims about the least likely bins. For $N \ge 4$ the cubic coefficient $2(\lfloor N/2 \rfloor - 1)(\lceil N/2 \rceil - 1)$ of S_N is positive, so $2u_N + (N-2)u_N^2 < S_N(u_N) = 1$, hence $u_N < u^{\mathrm{adj}}$ and $p_N > p_N^{\mathrm{adj}}$. At p_N the middle ties with the endpoints, which gives the global maximum, and bins $1, N-1$ are below the endpoints, which gives the global minimum. For $N = 2, 3$ we have $S_N = S_{1,N-1}$, and symmetry gives the ties. □

So $1 - p_N^{\mathrm{adj}} \sim N^{-1/2}$, much larger than $1 - p_N \sim (\log N)/N$. Away from the ends, the single letter + of bin 1 needs two switches (one into it and one out of it), so $S_{1,N-1} \approx Nu^2$ reaches 1 only at $u \approx N^{-1/2}$. (At $p = 0$ and $p = 1$ there are also ties between bins of probability 0.)

4.2 A Bessel estimate uniform in the bin

Words in bin a with $r+1$ runs of each letter have $2r+1$ switches. Their contribution to $S_{a,b}$ is about $2u(abu^2)^r/(r!)^2$. So the natural variable is abu^2, and a bin near the edge acts like the middle of a shorter row with $\sqrt{ab}$ in place of $N/2$. Put

$$s=\sqrt{ab},\quad A=\frac{ab}{a+b},\quad c=\frac{a+b}{2\sqrt{ab}},\quad \eta=\frac{1}{c},\quad x=su,\quad \mathcal{J}_c(x)=I_0(2x)+cI_1(2x),$$

and $\widehat{S}_{a,b}(u)=2u\mathcal{J}_c(su)$. Here $c\ge 1$ and $0<\eta\le 1$, with $\eta=1$ exactly when $a=b$. This Bessel form is classical. The fixed-scale law of Herrmann and Vallois [10, Remark 3.4] with a fair start has atoms $e^{-\tau}/2$ at ±1 and density $\frac{1}{2}\tau e^{-\tau}\{I_0(\tau\rho)+I_1(\tau\rho)/\rho\}$ at magnetization per spin m, where $\rho=\sqrt{1-m^2}$. At $m=(a-b)/N$ we have $\rho=2s/N$, and with τ replaced by Nu, the density times the bin width $2/N$, divided by an atom, is $\widehat{S}_{a,b}(u)$. The next theorem bounds how far the exact ratio is from it, at every N.

Theorem 4.3 (uniform estimate for every bin)**.** *For all positive integers a,b and all $u>0$,*

$$0\le 1-\frac{S_{a,b}(u)}{\widehat{S}_{a,b}(u)}\le\frac{(a+b)u^2}{2}+u=\frac{x^2}{2A}+\frac{x}{s}. \tag{14}$$

In particular, for fixed N and u the bound does not depend on the bin. The inequality still holds when the right side exceeds 1, but then it gives no lower bound on $S_{a,b}$.

When u is about L/N, the bound is $O(L^2/N)$ for every bin at once. The proof follows that of Theorem 3.3(i), with the two letters weighted by a and b.

Proof. Let $U_r=U_r(a)$ and $V_r=U_r(b)$ as in Lemma 3.2(b), and note that $1/a+1/b=1/A$. Lemma 3.2(a), applied to the $2r$ clipped factors of U_rV_r, gives

$$0\le 1-U_rV_r\le\frac{r(r+1)}{2A}. \tag{15}$$

Lemma 3.2(a) gives $1-U_{r+1}V_r\le\frac{(r+1)(r+2)}{2a}+\frac{r(r+1)}{2b}$, and the same with a and b exchanged for U_rV_{r+1}. So the weighted average $E_r=(aU_{r+1}V_r+bU_rV_{r+1})/(a+b)$ satisfies

$$\begin{aligned}0\le 1-E_r&\le\frac{(r+1)(r+2)}{a+b}+\frac{r(r+1)}{2(a+b)}\Big(\frac{a}{b}+\frac{b}{a}\Big)\\&=\frac{r(r+1)}{2A}+\frac{2(r+1)}{a+b}.\end{aligned} \tag{16}$$

Since $\binom{a-1}{r}=a^rU_r/r!$, equation (13) becomes

$$S_{a,b}(u)=2u\Big[\sum_{r\ge0}\frac{x^{2r}}{(r!)^2}U_rV_r+c\sum_{r\ge0}\frac{x^{2r+1}}{r!(r+1)!}E_r\Big], \tag{17}$$

and dropping U_rV_r and E_r gives $\widehat{S}_{a,b}(u)$. So by Lemma 3.2(c), with $I_j=I_j(2x)$, $(\widehat{S}_{a,b}-S_{a,b})/(2u)$ lies between 0 and

$$\frac{1}{2A}(x^2I_0+xI_1+cx^2I_1)+\frac{2c\,xI_0}{a+b}=\Big(\frac{x^2}{2A}+\frac{x}{s}\Big)\mathcal{J}_c(x),$$

since $c/s=1/(2A)$ and $2c/(a+b)=1/s$. Divide by $\widehat{S}_{a,b}/(2u)=\mathcal{J}_c(x)$. □

In Ising terms the theorem is a bound on the partition function of the fixed-magnetization ensemble, at every N, every coupling and every magnetization $|\mu| < N$.

Corollary 4.4 (fixed-magnetization weight)**.** *Let $|\mu| < N$ with $\mu \equiv N \pmod 2$, and let $Z_N(\mu)$ be the total weight of the configurations of the free chain with magnetization μ, as in Section 1.2. Put $\rho = \sqrt{1-\mu^2/N^2}$ and $v = Ne^{-2J}\rho$. Then for every real J,*

$$Z_N(\mu) = 2e^{(N-3)J}\Big(I_0(v) + \frac{I_1(v)}{\rho}\Big)(1-\theta), \qquad 0 \le \theta \le \frac{N}{2}e^{-4J} + e^{-2J}.$$

Proof. Put $a = (N+\mu)/2$ and $b = (N-\mu)/2$. Then $Z_N(\mu) = e^{(N-1)J}S_{a,b}(u)$ with $u = e^{-2J}$, and $2s = 2\sqrt{ab} = N\rho$, so $2su = v$ and $c = 1/\rho$. Now $\widehat{S}_{a,b}(u) = 2u(I_0(v) + I_1(v)/\rho)$ and $2ue^{(N-1)J} = 2e^{(N-3)J}$, and Theorem 4.3 bounds θ. □

Remark 4.5 (the sign of the first correction). Theorem 4.3 shows that the exact weight lies below its Bessel form, so the first correction to the Bessel form is negative. Its size is of the order of the bound, since the clipped products U_rV_r fall short of 1 by about $r(r+1)/(2A)$. Dantchev, Tonchev and Rudnick [4, Appendix D, Eq. (D.6)] give the correction to the Bessel form of ${}_2F_1(-a,-b;1;z)$ with the opposite sign. Their Eq. (D.4) writes $\binom{a}{n}n!$ as $(a+n-1)!/(a-1)!$, while it is the falling factorial $a!/(a-n)!$, and with the falling factorial the sign of the correction is reversed. For example, with their parameters $a = b = 200$ and $abz = 25$, the exact ${}_2F_1(-a,-b;1;z)$ is 9.5 percent below $I_0(2\sqrt{abz})$, while their Eq. (D.6) is 10.1 percent above it and the same formula with the opposite sign is within 0.8 percent of the exact value.

To turn Theorem 4.3 into a bound on the crossing, let $x_{a,b} = su_{a,b}$ and let $\hat{x}_{a,b}$ be the positive root of

$$G_\eta(\hat{x}_{a,b}) = A, \qquad G_\eta(x) = x\{I_1(2x) + \eta I_0(2x)\}. \tag{18}$$

Since $c\eta = 1$ and $2c/s = 1/A$, we have $\widehat{S}_{a,b}(u) = G_\eta(su)/A$. Differentiating term by term gives

$$G_\eta'(x) - 2G_\eta(x) = 2x(1-\eta)\{I_0(2x) - I_1(2x)\} + \eta I_0(2x) \ge 0, \tag{19}$$

since $I_0(2x) - I_1(2x) = \frac{1}{\pi}\int_0^\pi e^{2x\cos t}(1-\cos t)\,dt > 0$ by [12, Eq. 10.32.3]. So $(\log G_\eta)' \ge 2$ for $x > 0$ and $0 \le \eta \le 1$, and $\hat{x}_{a,b}$ is unique.

Corollary 4.6 (quantitative crossing bound)**.** *We have $x_{a,b} \ge \hat{x}_{a,b}$. If*

$$\delta_{a,b} = \frac{x_{a,b}^2}{2A} + \frac{x_{a,b}}{s} < 1,$$

then

$$0 \le x_{a,b} - \hat{x}_{a,b} \le -\tfrac{1}{2}\log(1-\delta_{a,b}). \tag{20}$$

Proof. At the crossing $S_{a,b} = 1$, so Theorem 4.3 gives $A \le G_\eta(x_{a,b}) \le A/(1-\delta_{a,b})$. As $G_\eta(\hat{x}_{a,b}) = A$, this gives $x_{a,b} \ge \hat{x}_{a,b}$, and integrating $(\log G_\eta)' \ge 2$ from $\hat{x}_{a,b}$ to $x_{a,b}$ gives (20). This is the argument of Theorem 3.3(ii). For even N and $a = b$ the two statements coincide, since then $\eta = 1$, $s = h$ and $2G_1 = F$, so $\hat{x}_{a,b} = x_B$. □

Remark 4.7 (Jacobi polynomials). The run sums of this paper can be written in terms of Jacobi polynomials, and their Bessel and Laguerre forms are limits of Jacobi polynomials near the point 1. For $a \le b$ and $0 \le t < 1$,

$$\sum_{r\ge 0}\binom{a-1}{r}\binom{b-1}{r}t^r = {}_2F_1(1-a,1-b;1;t) = (1-t)^{a-1}P_{a-1}^{(0,\,b-a)}\Big(\frac{1+t}{1-t}\Big),$$

and for $b > a$ and $0 \le u < 1$,

$$\sum_{r\ge 0} \binom{a-1}{r}\binom{b-1}{r+1} u^{2r+2} = \frac{(b-1)u^2}{a}(1-u^2)^{a-1} P_{a-1}^{(1,b-a-1)}\Big(\frac{1+u^2}{1-u^2}\Big).$$

For fixed $\alpha, \beta > -1$, the Mehler–Heine limit [12, Eq. 18.11.5] holds uniformly on bounded sets of complex arguments [20, Theorem 8.1.1]. Taken at the argument iw, it gives $n^{-\alpha} P_n^{(\alpha,\beta)}(1 + w^2/(2n^2)) \to (w/2)^{-\alpha} I_\alpha(w)$ as $n \to \infty$, uniformly for bounded $w \ge 0$, with the value at $w = 0$ taken by continuity. Elliott [21, Section 3, Eq. (3.10)] gives formal modified-Bessel expansions in the plane cut along $(-\infty, 1]$, away from -1, for fixed parameters with $\alpha \ge 0$. They apply on the real axis to the right of 1, where the arguments above lie, but he defers truncation-error bounds to a proposed later paper [21, Section 1]. For fixed a and $y \ge 0$, put $u^2 = y/b$ and $\beta = b - a - 1$. The argument of the second Jacobi polynomial is $1 - 2\tilde{x}/\beta$, where $\tilde{x} = -\beta y/(b - y) \to -y$ as $b \to \infty$. The fixed-degree Jacobi–Laguerre limit [12, Eq. 18.7.21] is locally uniform in its argument, by coefficient convergence of polynomials of fixed degree. It makes the last sum tend to $(y/a) L_{a-1}^{(1)}(-y) = H_a(y)$ of (25). The odd-run contribution, including its factor $2u$, and the other even-run contribution are respectively $O_a(b^{-1/2})$ and $O_a(b^{-1})$, uniformly for y in compact sets. Temme [22, Section 4.2] also compares Jacobi with Laguerre polynomials, and Gil, Segura and Temme [23, Section 2 and Remark 1] expand $P_n^{(\alpha,\beta)}(1 - 2x/\beta)$ in inverse powers of β with Laguerre coefficients, uniformly for bounded n, α and x (complex x included). This covers the fixed degree $a - 1$ of Section 4.4, while Theorem 4.9 lets the degree grow, uniformly in $b \ge a$. Also, Theorem 4.10 is about the root of $H_a(y) = 1$, a level of the Laguerre polynomial at the negative argument $-y$, and not about its zeros. Here $b - a$ can be as large as $N - 2$, and Theorem 4.3 gives an explicit relative error at every N.

4.3 Every growing distance from an edge

The Bessel root satisfies an equation of the form $z + \frac{1}{2}\log z = \ell + C + B/z + \dots$. The next lemma inverts such equations.

Lemma 4.8 (inverting $z + \frac{1}{2}\log z$)**.** *Fix $K > 0$. There is a constant C_K with the following property. Let $z \ge 1$ and $\ell \ge 2$, and suppose that $z + \frac{1}{2}\log z = \ell + C + B/z + \omega$ with $|C|, |B| \le K$ and $|\omega| \le 1$. Then*

$$\Big| z - \ell + \tfrac{1}{2}\log\ell - C - \frac{\frac{1}{4}\log\ell - \frac{1}{2}C + B}{\ell} \Big| \le C_K\Big(\frac{(\log\ell)^2}{\ell^2} + |\omega|\Big).$$

Proof. Put $w = z - \ell$. As $z \ge 1$, $\log z \ge 0$ and $|B/z| \le K$, so $z \le \ell + 2K + 1$ and $|w| \le 2K + 1 + \frac{1}{2}\log(\ell + 2K + 1) = O_K(\log\ell)$. This proves the claim for ℓ bounded in terms of K, since then both sides are $O_K(1)$ and $(\log\ell)^2/\ell^2$ is bounded below. For larger ℓ we have $|w| \le \ell/2$, so $\log z = \log\ell + w/\ell + O((w/\ell)^2)$ and $B/z = B/\ell + O_K(|w|/\ell^2)$, and $w = C - \frac{1}{2}\log\ell - w/(2\ell) + B/\ell + \omega + O_K((\log\ell)^2/\ell^2)$. This gives first $w = d + O_K(\log\ell/\ell + |\omega|)$ with $d = C - \frac{1}{2}\log\ell$, and then, putting this back into the term $w/(2\ell)$, $w = d + (B - d/2)/\ell + O_K((\log\ell)^2/\ell^2 + |\omega|)$. □

Theorem 4.9 (uniform crossing expansion)**.** *Let $1 \le a \le b$, let A and η be as in Section 4.2, and put*

$$\ell = \log A, \qquad C(\eta) = \tfrac{1}{2}\log(8\pi) - \log(1+\eta), \qquad B(\eta) = \frac{3-\eta}{8(1+\eta)}.$$

As $a \to \infty$, uniformly over $b \geq a$,

$$2\sqrt{ab}\, u_{a,b} = \ell - \tfrac{1}{2}\log\ell + C(\eta) + \frac{\frac{1}{4}\log\ell - \frac{1}{2}C(\eta) + B(\eta)}{\ell} + O\left(\frac{(\log\ell)^2}{\ell^2} + \frac{(\log a)^2}{a}\right). \tag{21}$$

That is, there are a_0 and K' such that the error term is at most K' times the bracket for all $a \geq a_0$ and all $b \geq a$. The same holds with $1 - p_{a,b}$ in place of $u_{a,b}$.

Here $A = ab/(a+b)$ is half the harmonic mean of a and b. There is no condition on how a compares with N, so the theorem covers a bin at distance $\sqrt{N}$ from the edge as well as the middle. The proof uses Corollary 4.6 to pass to the Bessel root $\hat{x}_{a,b}$. The large-argument form of I_0 and I_1 then turns (18) into an equation that Lemma 4.8 solves.

Proof. Here $a/2 \leq A \leq a$, $s \geq a$ and $0 < \eta \leq 1$.

Step 1. The crossing is at $x_{a,b} \leq \log a$. At $x = \log a$ the bound of Theorem 4.3 is at most $2(\log a)^2/a$, while $\widehat{S}_{a,b} = G_\eta(x)/A \geq xI_1(2x)/a \to \infty$, both uniformly in b. So $S_{a,b} > 1$ there, and since $S_{a,b}$ increases, $x_{a,b} \leq \log a$ for all $a \geq a_0$ and all $b \geq a$. Then $\delta_{a,b} \leq 2(\log a)^2/a$, and Corollary 4.6 gives

$$x_{a,b} - \hat{x}_{a,b} = O((\log a)^2/a). \tag{22}$$

Step 2. The Bessel root. Write $z = 2\hat{x}_{a,b}$, so $z \to \infty$ as $A \to \infty$, uniformly in η. By the large-argument form of I_0 and I_1 [12, Eq. 10.40.1], $I_1(z)+\eta I_0(z) = (1+\eta)e^z(2\pi z)^{-1/2}(1-B(\eta)/z+O(z^{-2}))$ uniformly in $0 \leq \eta \leq 1$. Taking logarithms in (18) gives $z + \frac{1}{2}\log z = \ell + C(\eta) + B(\eta)/z + O(z^{-2})$. As $|C(\eta)| \leq 2$ and $0 < B(\eta) \leq \frac{3}{8}$, Lemma 4.8 with $K = 2$ expands z.

Step 3. Back to the crossing. By (22) the expansion moves to $2x_{a,b} = 2\sqrt{ab}\, u_{a,b}$. Finally $2s(u - u/(1+u)) \leq 2x^2/s = O((\log a)^2/a)$ at $u = u_{a,b}$, which gives the same expansion for $1 - p_{a,b} = u_{a,b}/(1+u_{a,b})$. □

Hence, uniformly in b,

$$1 - p_{a,b} \sim \frac{\log a}{2\sqrt{ab}} \qquad (a \to \infty,\ b \geq a). \tag{23}$$

If $a \to \infty$ with $a = o(b)$, then $\eta \to 0$ and $\log A = \log a + o(1)$, so

$$2\sqrt{ab}(1 - p_{a,b}) = \log a - \tfrac{1}{2}\log\log a + \tfrac{1}{2}\log(8\pi) + o(1). \tag{24}$$

If $a/N \to \alpha \in (0, \frac{1}{2}]$, however slowly, then

$$2\sqrt{a(N-a)}(1 - p_{a,N-a}) = \log N - \tfrac{1}{2}\log\log N + \log\{\alpha(1-\alpha)\} + C(2\sqrt{\alpha(1-\alpha)}) + o(1).$$

At the center for even N, $a = b = N/2$ gives $A = N/4$ and $\eta = 1$, and changing from $\log A$ to $\log N$ recovers (11).

4.4 Fixed bins and the Laguerre edge law

In Ising terms, bin a with a fixed is magnetization $2a - N$, a few up spins in a sea of down spins, and we ask when it is as likely as full order. Let $b \to \infty$. A run of letters + away from the ends

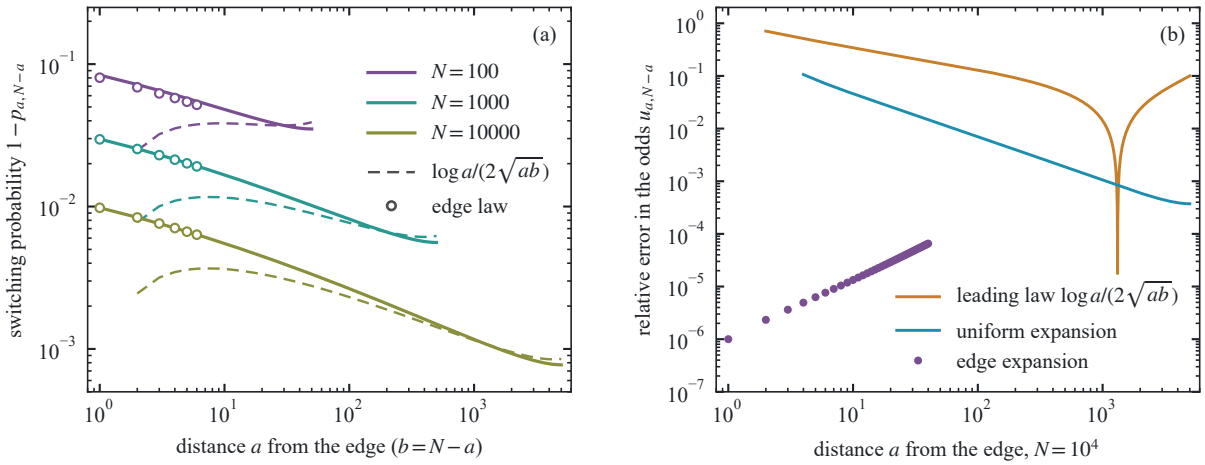


Figure 3: (a) The switching probability $1-p_{a,N-a}$ at the crossing of each bin, from the exact run polynomial, against its distance a from the nearer edge, for $N = 100$, 1000 and 10^4. The dashed lines are the leading law (23). It is asymptotic in a, so it departs from the exact curves near the edge. The circles are the first two terms of the edge law (27) for $a \le 6$. The right end of each line is the central crossing $1-p_N$. (b) For $N = 10^4$, the absolute relative error in the odds $u_{a,N-a}$ of the leading law $(\log a)/(2\sqrt{ab})$ with $b = N-a$, of the main terms of the uniform expansion (21) for $a \ge 4$, and of the first three terms of the edge expansion (26) for $a \le 40$. The dip near $a = 1300$ is where the error of the leading law changes sign.

needs two switches, which gives a factor u^2, and it can go in about b places. So the right variable is $y = bu^2$. Define

$$H_a(y) = \sum_{r=0}^{a-1} \binom{a-1}{r} \frac{y^{r+1}}{(r+1)!} = \frac{y}{a} L_{a-1}^{(1)}(-y), \tag{25}$$

with the standard Laguerre polynomial $L_{a-1}^{(1)}$. Let y_a be the positive root of $H_a(y_a) = 1$ and $c_a = \sqrt{y_a}$. Then $y_1 = 1$, $y_2 = \sqrt{3}-1$, and y_a decreases strictly in a, by coefficient comparison.

Theorem 4.10 (fixed-edge expansion)**.** *For each fixed positive integer a, as $b \to \infty$,*

$$u_{a,b} = \frac{c_a}{\sqrt{b}} - \frac{1}{b} + \frac{\gamma_a}{b^{3/2}} + O_a(b^{-2}), \tag{26}$$

where

$$\gamma_a = \frac{1 + y_a + (y_a + y_a^2/2)H_a''(y_a)/H_a'(y_a)}{2c_a}.$$

Thus the second coefficient in the odds expansion is -1 for every fixed bin. For the switching probability,

$$1 - p_{a,b} = \frac{c_a}{\sqrt{b}} - \frac{1+c_a^2}{b} + O_a(b^{-3/2}). \tag{27}$$

For bin 1, where $y_1 = 1$ and $H_1'' = 0$, this gives $u_{1,b} = b^{-1/2} - b^{-1} + b^{-3/2} + \dots$, the expansion of the exact root $1/(1+\sqrt{b})$ of Proposition 4.2.

Proof. Put $\theta = b^{-1/2}$, $y = bu^2$ and $V_r = U_r(b)$. For $b \ge a$, (13) reads

$$S_{a,b}(u) = 2\sqrt{y/b}\sum_r \binom{a-1}{r}\frac{y^r}{r!}V_r + \frac{y}{b}\sum_r \binom{a-1}{r+1}\frac{y^r}{r!}V_r + y\sum_r \binom{a-1}{r}\frac{y^r}{(r+1)!}V_{r+1}. \tag{28}$$

These sums have at most a terms, and with $V_r = 1$ they are $H'_a(y)$, $H''_a(y)$ and $H_a(y)$. Expanding $V_r = 1 - r(r+1)/(2b) + O_a(b^{-2})$ and using $\sum_r (r+1)(r+2)\binom{a-1}{r}\frac{y^{r+1}}{(r+1)!} = y^2H''_a(y) + 2yH'_a(y)$ gives, uniformly near $y = y_a$,

$$S_{a,b}(u) = H_a(y) + 2\sqrt{y}\,H'_a(y)\theta + \{(y - y^2/2)H''_a(y) - yH'_a(y)\}\theta^2 + O_a(\theta^3). \tag{29}$$

With $b = \theta^{-2}$ the right side of (28) is analytic in (θ, y) near $(0, y_a)$, and $H'_a(y_a) > 0$. So the implicit-function theorem gives an analytic solution $y(\theta) = y_a + d_1\theta + d_2\theta^2 + O_a(\theta^3)$ of $S_{a,b} = 1$, and by the uniqueness of the crossing it is the value of y at $u = u_{a,b}$. Matching coefficients in (29) gives $d_1 = -2c_a$ and $d_2 = 2 + y_a + (y_a + y_a^2/2)H''_a(y_a)/H'_a(y_a)$. Expanding $u = \theta\sqrt{y}$ and then $u/(1+u)$ proves both expansions. □

The next theorem shows that for large a the edge constant $2\sqrt{ay_a}$, the limit of $2\sqrt{ab}\,u_{a,b}$ as $b \to \infty$, has the expansion of the growing-distance law (24).

Theorem 4.11 (matching the edge and Bessel regimes)**.** *For every positive integer a and every $x > 0$,*

$$0 \le 1 - \frac{aH_a(x^2/a)}{xI_1(2x)} \le \frac{x^2}{2a}. \tag{30}$$

Consequently, with $\ell = \log a$ and with $C(0) = \frac{1}{2}\log(8\pi)$ from Theorem 4.9, as $a \to \infty$,

$$2\sqrt{ay_a} = \ell - \tfrac{1}{2}\log\ell + C(0) + \frac{\frac{1}{4}\log\ell - \frac{1}{2}C(0) + 3/8}{\ell} + O\left(\frac{(\log\ell)^2}{\ell^2} + \frac{(\log a)^2}{a}\right). \tag{31}$$

Proof. With $U_r = U_r(a)$, $aH_a(x^2/a) = x\sum_{r\ge0} U_r x^{2r+1}/(r!(r+1)!)$, so Lemma 3.2(b) and the second identity of Lemma 3.2(c) give (30). Put $\omega_a = \sqrt{ay_a}$, so $aH_a(\omega_a^2/a) = a$, and $\omega_a < \log a$ for large a by (30). So $a \le G_0(\omega_a) \le a/(1-\omega_a^2/(2a))$, and as in Corollary 4.6, $\omega_a - \hat\omega_a = O((\log a)^2/a)$, where $G_0(\hat\omega_a) = a$. So $z = 2\hat\omega_a$ satisfies $z + \frac{1}{2}\log z = \ell + C(0) + 3/(8z) + O(z^{-2})$ by [12, Eq. 10.40.1], as in the proof of Theorem 4.9 with $\eta = 0$ and $A = a$, and Lemma 4.8 gives the expansion. □

Figure 3 compares these laws with the exact crossings.

4.5 The moving boundary layer

The crossings also give the shape of a whole row at odds $u = v_N > 0$. Between the central crossing (v_N about $(\log N)/N$) and the adjacent one (v_N about $N^{-1/2}$), the middle is above the endpoints and a layer of bins next to each edge is below them. (This is a layer of bins near each end of the row, that is, in the magnetization, not a region of the chain.) The width of this layer is

$$\mathcal{L}_N = \#\{1 \le k \le \lfloor N/2\rfloor : P_N(k) < P_N(0)\}.$$

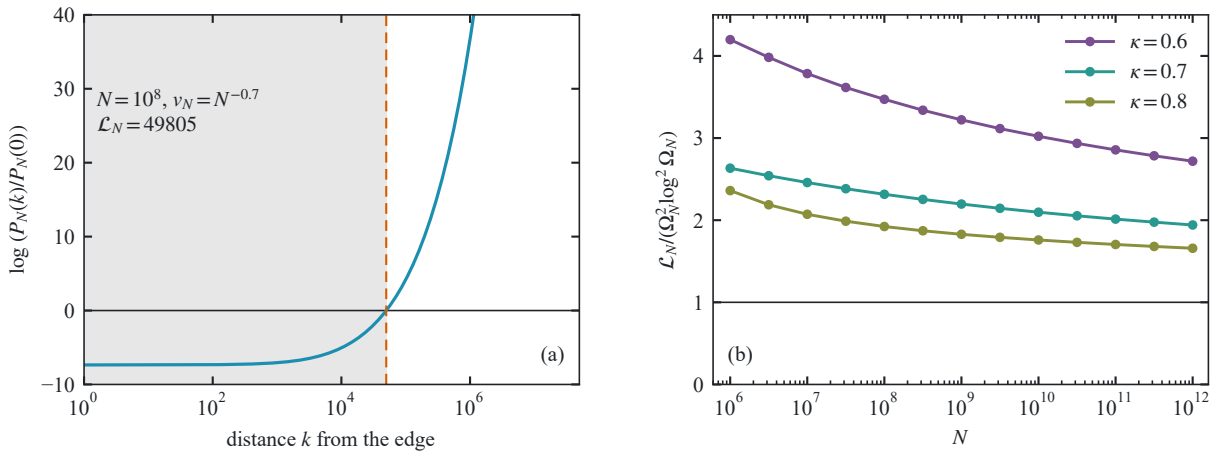


Figure 4: (a) The logarithm $\log(P_N(k)/P_N(0))$ of the ratio of each bin to an endpoint, from the exact run polynomial, against the distance k from the nearer edge, at $N = 10^8$ and odds $v_N = N^{-0.7}$. The shaded bins are below the endpoint height, and there are $\mathcal{L}_N = 49805$ of them in the left half. (b) The width $\mathcal{L}_N$ divided by its leading term $\Omega_N^2 \log^2 \Omega_N$ of (32), for $v_N = N^{-\kappa}$ ($\kappa = 0.6$, 0.7 and 0.8) and $10^6 \le N \le 10^{12}$. The ratios decrease toward 1 only slowly.

Corollary 4.12 (boundary-layer width)**.** *Suppose that, as* $N \to \infty$*,*

$$\Omega_N = \frac{1}{\sqrt{N}\, v_N} \longrightarrow \infty, \qquad \frac{\log \Omega_N}{N v_N} \longrightarrow 0.$$

Then

$$\mathcal{L}_N \sim \Omega_N^2 \log^2 \Omega_N = \frac{\log^2(1/(\sqrt{N}\, v_N))}{N v_N^2}. \tag{32}$$

In particular, if $v_N = N^{-\kappa}$ *with* $1/2 < \kappa < 1$*, then*

$$\mathcal{L}_N \sim (\kappa - \tfrac{1}{2})^2 N^{2\kappa - 1} \log^2 N.$$

By symmetry, the total number of interior bins below the endpoint height is $2\mathcal{L}_N$ *for all sufficiently large* N*.*

Proof. Set $d_N = \Omega_N^2 \log^2 \Omega_N$. Then $d_N \to \infty$ and $d_N/N = (\log \Omega_N/(N v_N))^2 \to 0$. Fix $\epsilon \in (0,1)$ and integers $a_\pm = (1 \pm \epsilon) d_N + O(1)$. Then $\log a_\pm \sim 2 \log \Omega_N$ and $\sqrt{a_\pm (N - a_\pm)}\, v_N \sim \sqrt{1 \pm \epsilon}\, \log \Omega_N$, so (23) gives $u_{a_\pm, N - a_\pm}/v_N \to 1/\sqrt{1 \pm \epsilon}$. Bin k is below the endpoint exactly when $v_N < u_{k,N-k}$, and $u_{k,N-k}$ decreases strictly for $k \le N/2$ by Proposition 4.1. So $a_- \le \mathcal{L}_N < a_+$ for large N, and we let $\epsilon \downarrow 0$. Finally $\mathcal{L}_N = o(N)$, so the reflected intervals are disjoint. □

Figure 4 shows one row inside this regime and the width $\mathcal{L}_N$ for three exponents κ. If $\mathcal{L}_N$ is instead divided by the distance a at which the main terms of (21), divided by $2\sqrt{a(N-a)}$, equal v_N, the ratio is within 0.013 of 1 for every N in panel (b).

5 Periodic boundaries and the crossing window

5.1 Periodic boundaries

We now close the sequence of steps into a loop, so that the last step also interacts with the first. With this closing bond the spins form the periodic Ising chain, with weight proportional

to $\exp(J\sum_{i=1}^{N}\sigma_i\sigma_{i+1})$, $\sigma_{N+1}=\sigma_1$ and again $u=e^{-2J}$. Antal, Droz and Rácz [2] give its magnetization law at a fixed switching scale τ, and Dantchev and Rudnick [3] give exact coefficients. We find how much the closing bond moves the central crossing, and then describe the row near the crossing for both boundary conditions.

Write $S^{\mathrm{F}}_{N,k}=S_{k,N-k}$ and $S^{\mathrm{P}}_{N,k}$ for the bin-to-endpoint ratios for free and periodic ends. Put $a=k$, $b=N-k$ and $s=\sqrt{ab}$. Counting cyclic words by their runs gives

$$S^{\mathrm{P}}_{N,k}(u)=\sum_{r=1}^{\min(a,b)}\frac{N}{r}\binom{a-1}{r-1}\binom{b-1}{r-1}u^{2r}. \tag{33}$$

To see this, note that a cyclic word with a letters $+$ and b letters $-$ switches an even number $2r$ of times and has r runs of each letter. Choose compositions of a and of b into r positive runs and a position for the start of a distinguished run of letters $+$, and divide by the r choices of that run. This is the count of [2, Eq. (30)] and [3, Eq. (2.9)]. The normalization of the periodic measure cancels in the ratio. In particular $S^{\mathrm{P}}_{N,1}=Nu^2$, which gives the adjacent crossing $J=(\log N)/4$ of Stepanyan et al. [6]. Every interior ratio increases strictly from 0 to ∞ in $u>0$, and the ratios increase strictly toward the middle, apart from the symmetric middle tie, by the coefficient comparison in the proof of Proposition 4.1, applied to (33).

Theorem 5.1 (periodic relative error)**.** *For $N\geq 2$, $1\leq k\leq N-1$, and $u>0$, define*

$$\widehat{S}^{\mathrm{P}}_{N,k}(u)=\frac{Nu}{\sqrt{k(N-k)}}I_1\Bigl(2u\sqrt{k(N-k)}\Bigr).$$

Then

$$0\leq 1-\frac{S^{\mathrm{P}}_{N,k}(u)}{\widehat{S}^{\mathrm{P}}_{N,k}(u)}\leq\frac{Nu^2}{2}. \tag{34}$$

So for periodic ends the ratio of every bin to an endpoint is close to a single Bessel function I_1, with the same kind of error as in Theorem 4.3.

Proof. Put $x=su$, $A=ab/N$, $U_r=U_r(a)$ and $V_r=U_r(b)$. Shifting the index in (33) gives $S^{\mathrm{P}}_{N,k}(u)=(Nu/s)\sum_{r\geq 0}U_rV_rx^{2r+1}/(r!(r+1)!)$, so the ratio to $\widehat{S}^{\mathrm{P}}_{N,k}$ is an average of the U_rV_r with weights $x^{2r+1}/(r!(r+1)!)$, whose sum is $I_1(2x)$. By (15), $0\leq 1-U_rV_r\leq r(r+1)/(2A)$, and the second identity of Lemma 3.2(c) bounds the deficit by $x^2/(2A)=Nu^2/2$. □

A cyclic word switches an even number of times, so only I_1 appears in the periodic approximation. The free words with an odd number of switches start and end on different letters, and they give the extra I_0 term of the free one. As $I_0\approx I_1$ for large arguments, the free ratio is about twice the periodic one near the crossing. Both grow like e^{Nu}, so the periodic crossing needs Nu about $\log 2$ larger. Antal, Droz and Rácz [2, Eqs. (37) and (47)] work at a fixed value of $\tau=Nq$, their scaling variable (Section 1.2). There the endpoint atoms are $1/(2\cosh\tau)$ (periodic) and $e^{-\tau}/2$ (free), which again differ by a factor of about 2 for large τ, while their densities at zero magnetization agree to leading order as $\tau\to\infty$.

Theorem 5.2 (periodic crossing and boundary shift)**.** *Let u_N^{P} be the positive root of $S_{N,m_N}^{\mathrm{P}}(u) = 1$, let $p_N^{\mathrm{P}} = 1/(1+u_N^{\mathrm{P}})$, and let $u_N^{\mathrm{F}} = u_N$ be the free central root. Put $c_{\mathrm{P}} = \frac{1}{2}\log(\pi/2)$. As $N \to \infty$,*

$$Nu_N^{\mathrm{P}} = L - \tfrac{1}{2}\log L + c_{\mathrm{P}} + \frac{\frac{1}{4}\log L - \frac{1}{2}c_{\mathrm{P}} + \frac{3}{8}}{L} + O\!\left(\frac{(\log L)^2}{L^2}\right), \tag{35}$$

$$N(u_N^{\mathrm{P}} - u_N^{\mathrm{F}}) = \log 2 + \frac{\frac{1}{4} - \frac{1}{2}\log 2}{L} + O\!\left(\frac{(\log L)^2}{L^2}\right). \tag{36}$$

Replacing Nu_N^{P} by $N(1-p_N^{\mathrm{P}})$ preserves (35). At the periodic crossing the coupling $J_N^{\mathrm{P}} = -\frac{1}{2}\log u_N^{\mathrm{P}}$ satisfies

$$J_N^{\mathrm{P}} = \tfrac{1}{2}(\log N - \log\log N) + \frac{\frac{1}{4}\log\log N - \frac{1}{2}c_{\mathrm{P}}}{\log N} + O\!\left(\frac{(\log\log N)^2}{(\log N)^2}\right). \tag{37}$$

Since $c_{\mathrm{P}} = c_0 + \log 2$, the periodic expansion is the free one (11) with c_0 replaced by $c_0 + \log 2$, and with $\frac{3}{8}$ in place of $\frac{1}{8}$ in the $1/L$ term.

Proof. Step 1. The first order. At the middle $2s = N + O(1/N)$. At $u = (1 \pm \theta)L/N$ with fixed $\theta \in (0,1)$ the bound of Theorem 5.1 tends to 0, and the large-argument form of I_1 makes $\widehat{S}^{\mathrm{P}}$ tend to ∞ with the plus sign and to 0 with the minus sign. Since S_{N,m_N}^{P} increases in u, we get $Nu_N^{\mathrm{P}} \sim L$.

Step 2. The crossing equation. Write $z = Nu_N^{\mathrm{P}}$. At the root $S^{\mathrm{P}} = 1$, so $\log \widehat{S}^{\mathrm{P}} = O(L^2/N)$ by (34). Replacing the argument $2su$ by z, and the factor Nu/s by $2z/N$, changes $\log \widehat{S}^{\mathrm{P}}$ by $O(L/N^2)$. Here $(\log I_1)' = I_0/I_1 - 1/z = O(1)$ for large positive arguments by [12, Eqs. 10.29.2, 10.40.1]. Also $I_1(z) = e^z(2\pi z)^{-1/2}(1 - 3/(8z) + O(z^{-2}))$ [12, Eq. 10.40.1]. So

$$z + \tfrac{1}{2}\log z = L + c_{\mathrm{P}} + \frac{3}{8z} + O(z^{-2} + L^2/N),$$

and Lemma 4.8 with $\ell = L$, $C = c_{\mathrm{P}}$ and $B = \frac{3}{8}$ gives (35). As $N(u - u/(1+u)) = O(L^2/N)$ for $Nu = O(L)$, it also holds for $N(1 - p_N^{\mathrm{P}})$.

Step 3. The shift and the coupling. By (2), $Nu_N = z_N + O(L^2/N)$. Subtracting (11), where $c_0 = c_{\mathrm{P}} - \log 2$ and the constant in the $1/L$ term is $\frac{1}{8}$, gives (36). Finally $J = -\frac{1}{2}\log u = \frac{1}{2}(L - \log z)$, and $\log z = \log L - (\frac{1}{2}\log L - c_{\mathrm{P}})/L + O((\log L)^2/L^2)$ gives (37). □

5.2 The crossing window

Before we return to the coupling (37) in Section 5.3, we describe the whole row near the crossing. The factor e^{-2y^2} in the next theorem has a simple heuristic. Moving $yN/\sqrt{\log N}$ away from the middle multiplies the Bessel argument by $2\sqrt{k(N-k)}/N \approx 1 - 2y^2/\log N$. The argument is about $\log N$, so it drops by about $2y^2$, and the ratio, which grows like the exponential of the argument, drops by about the factor e^{-2y^2}.

Theorem 5.3 (crossing window in the magnetization)**.** *Let $X \in \{\mathrm{F}, \mathrm{P}\}$ and let u_N^X be the exact central crossing in the odds under boundary X (so $u_N^{\mathrm{F}} = u_N$). For fixed real t, y set*

$$u = u_N^X + \frac{t}{N}, \qquad k_N(y) = \operatorname{nint}\!\left(\frac{N}{2} + \frac{yN}{\sqrt{\log N}}\right),$$

where either nearest-integer convention may be used. Then, as $N \to \infty$, uniformly on compact sets of (t, y),

$$S_{N,k_N(y)}^X(u_N^X + t/N) \longrightarrow e^{t-2y^2}. \tag{38}$$

If $M_N^X(t)$ is the number of interior bins with probability strictly above the endpoint probability at the odds $u_N^X + t/N$, then

$$\frac{\sqrt{\log N}}{N} M_N^X(t) \longrightarrow \sqrt{2\max(t,0)}. \tag{39}$$

So the crossing window has width about $1/N$ in u and about $N/\sqrt{\log N}$ bins in the magnetization, and inside it the ratio of each bin to an endpoint is a Gaussian bump times e^t. The fixed-τ limit laws of [2, 10] cannot give this, since here τ grows like $\log N$. The Gaussian itself is expected. At $m = 2y/\sqrt{\log N}$ the factor e^{-2y^2} is $e^{-m^2 \log N/2}$, the large-τ Gaussian $e^{-m^2\tau/2}$ of Antal, Droz and Rácz [2, Section IV.E, Eq. (51)] at $\tau \approx \log N$. Theorem 5.3 adds the comparison with the endpoint, centered at the exact crossing and uniform on compact sets.

Proof. Step 1. The profile. Put $\lambda = k/N - \frac{1}{2}$ and $\eta = 2\sqrt{k(N-k)}/N = \sqrt{1-4\lambda^2}$. For the stated rounding and bounded y,

$$\eta = 1 - \frac{2y^2}{L} + O\Big(L^{-2} + \frac{1}{N\sqrt{L}}\Big), \qquad N u_N^X = L + O(\log L).$$

So the Bessel argument $Nu\eta$ at (u,k) is $Nu_N^X + t - 2y^2 + o(1)$, uniformly on the compact set. By Theorems 4.3 and 5.1 both ratios are within relative error $O(L^2/N)$ of their Bessel forms, and $I_0(z)$ and $I_1(z)$ equal $e^z(2\pi z)^{-1/2}(1 + O(1/z))$ [12, Eq. 10.40.1], so the prefactors relative to their central values tend to 1 uniformly. The change of argument gives e^{t-2y^2}, and the exact central ratio is 1, which proves (38).

Step 2. The count. The interior ratios increase toward the middle, by Proposition 4.1 for $X = \mathrm{F}$ and by the coefficient comparison after (33) for $X = \mathrm{P}$. If $t \le 0$, the central ratio is at most 1, so no interior bin is above the endpoint. Let $t > 0$ and $0 < \theta < \sqrt{t/2}$. By Step 1, for large N the bins at scaled distance less than $\sqrt{t/2} - \theta$ are above the endpoint and those at distance $\sqrt{t/2} + \theta$ are below, and by monotonicity so are all farther bins. Count the bins in between, divide by $N/\sqrt{L}$ and let $\theta \downarrow 0$. Rounding changes the count by $O(1)$. □

Remark 5.4 (constrained free energies). Let $Z_N^X(\mu)$ be the sum of the Ising weights of the configurations with magnetization μ under boundary X, and $\mathcal{F}_N^X(\mu) = -\log Z_N^X(\mu)$ its dimensionless free energy, as in the fixed-magnetization ensemble of [3, 5]. The normalization cancels in the ratio of a bin to an endpoint, so $-\log S_{N,k}^X = \mathcal{F}_N^X(2k-N) - \mathcal{F}_N^X(N)$. On a compact set of (t,y) the limit e^{t-2y^2} in (38) is bounded below by a positive constant, so the convergence there also holds for logarithms, uniformly. Hence

$$\mathcal{F}_N^X(2k_N(y) - N) - \mathcal{F}_N^X(N) \longrightarrow 2y^2 - t$$

at the odds $u_N^X + t/N$. Thus, near the crossing the free energy of a fixed magnetization, measured from full magnetization, is a parabola in the scaled magnetization, lowered by t. This compares single bins. About $N/\sqrt{\log N}$ bins of the window are each comparable with an endpoint, and the probabilities sum to one, so each of them has probability at most of order $\sqrt{\log N}/N$ there. The limit therefore says nothing about the total weight of the nearly ordered configurations.

Figure 5 shows the exact ratios and counts for $N = 10^4$, 10^6 and 10^8.

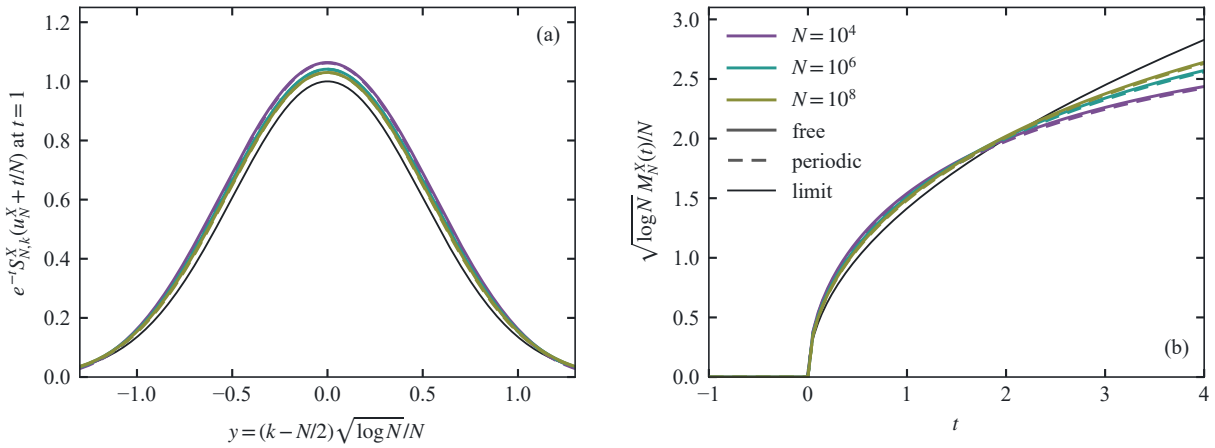


Figure 5: (a) The profile $e^{-t}S^X_{N,k}(u^X_N + t/N)$ at $t = 1$ against $y = (k - N/2)\sqrt{\log N}/N$, for free (solid) and periodic (dashed) ends and $N = 10^4$, 10^6 and 10^8, with the limit e^{-2y^2} of Theorem 5.3 (thin black line). (b) The scaled number $\sqrt{\log N}\, M^X_N(t)/N$ of interior bins above the endpoint height, against t, with the limit $\sqrt{2\max(t, 0)}$. All curves come from the exact run polynomials at the exact central crossings. The convergence is slow. At $t = 2$ and $y = 0$ the relative error of the profile is 0.12 at $N = 10^4$, 0.08 at $N = 10^6$ and 0.06 at $N = 10^8$, under either boundary.

5.3 The coupling at the crossing

We now turn to the coupling at the two central crossings, since this is how Stepanyan et al. [6] state the crossing. Recall the free coupling $J^{\mathrm{F}}_N = -\frac{1}{2}\log u_N$ and the correlation length $\xi(J) = -1/\log\tanh J$ of the infinite chain (Section 1.2), so that $\langle\sigma_0\sigma_r\rangle = e^{-r/\xi(J)}$ there, evaluated at the coupling of the crossing. The spin correlations in a chain of fixed magnetization behave differently, and can even become negative [24, Sections II and IV].

Corollary 5.5 (couplings and correlation lengths at the crossings)**.** *As $N \to \infty$,*

(a) The free coupling satisfies $J^{\mathrm{F}}_N = \frac{1}{2}(\log N - \log\log N) + \dfrac{\frac{1}{4}\log\log N - \frac{1}{2}c_0}{\log N} + O\Big(\dfrac{(\log\log N)^2}{(\log N)^2}\Big)$.

(b) The two couplings differ by $J^{\mathrm{F}}_N - J^{\mathrm{P}}_N = \dfrac{\log 2}{2\log N} + O\Big(\dfrac{\log\log N}{(\log N)^2}\Big)$.

(c) The correlation lengths satisfy $N/\xi(J^{\mathrm{F}}_N) = 2\log N - \log\log N + \log(\pi/8) + O(\log\log N/\log N)$, *and* $N/\xi(J^{\mathrm{P}}_N) = 2\log N - \log\log N + \log(\pi/2) + O(\log\log N/\log N)$.

All three parts are facts about Nu at the crossings, read on other scales. The coupling is $-\frac{1}{2}\log u$, so it sees Nu only through $\log(Nu)$. The shift of $\log 2$ in Nu between the two boundary conditions therefore becomes a shift of order $1/\log N$ in the coupling. The correlation length is $1/(2u)$ to leading order, so N/ξ is about $2Nu$.

Proof. (a) By (2), $u_N = z_N/(N - z_N)$, so $J^{\mathrm{F}}_N = \frac{1}{2}(L - \log z_N) + O(L/N)$. Taking logarithms in (11) gives $\log z_N = \log L - (\frac{1}{2}\log L - c_0)/L + O((\log L)^2/L^2)$, and (a) follows in the same way as (37) in Step 3 of the proof of Theorem 5.2.

(b) The difference is $J^{\mathrm{F}}_N - J^{\mathrm{P}}_N = \frac{1}{2}\log(u^{\mathrm{P}}_N/u^{\mathrm{F}}_N) = \frac{1}{2}\log(1 + \delta)$, where $\delta = N(u^{\mathrm{P}}_N - u^{\mathrm{F}}_N)/(Nu^{\mathrm{F}}_N)$. By (36) the numerator is $\log 2 + O(1/L)$. By (2) and Corollary 3.7(b) the denominator is $z_N + O(L^2/N) = L + O(\log L)$. So $\delta = (\log 2)/L + O(\log L/L^2)$, and $\frac{1}{2}\log(1 + \delta) = \delta/2 + O(\delta^2)$.

N	J_N^{P}	(37)	[6, (16)]	[6, figure 1(b)]	J_N^{F}	$J_N^{\mathrm{F}} - J_N^{\mathrm{P}}$
10	0.687	0.776	0.721	0.679	0.823	0.136
10^2	1.574	1.597	1.642	1.569	1.658	0.084
10^3	2.534	2.541	2.563	2.459	2.591	0.057
10^4	3.540	3.543	3.484	3.348	3.582	0.042
10^5	4.576	4.578	4.405	4.238	4.610	0.033
10^6	5.633	5.634	5.326	5.128	5.661	0.028

Table 2: The couplings J_N^{P} and J_N^{F} at the periodic and free central crossings, from the exact run polynomials and again from the transfer matrix (the two agree to 10^{-12}), against the main terms of (37) and the two approximations of Stepanyan et al. [6] (their Eq. (16) and the line of their Fig. 1(b)). By Corollary 5.5(b) the last column is $(\log 2)/(2\log N)$ to leading order, which is 0.030 at $N = 10^5$. Differences are computed before rounding.

(c) With $u = e^{-2J}$ we have $\tanh J = (1-u)/(1+u)$, so $N/\xi(J) = N \log \frac{1+u}{1-u} = 2Nu + O(Nu^3)$ for $0 < u \le \frac{1}{2}$. At either crossing $Nu^3 = O(L^3/N^2)$. It remains to insert $Nu_N = z_N + O(L^2/N) = L - \frac{1}{2}\log L + c_0 + O(\log L/L)$, from Corollary 3.7(b), and the expansion (35) of Nu_N^{P}, and to recall that $2c_0 = \log(\pi/8)$ and $2c_{\mathrm{P}} = \log(\pi/2)$. □

So at either crossing the correlation length is about $N/(2\log N)$. The chain is ordered over long stretches, but each stretch is a vanishing fraction of the chain. In the fixed-τ regime of [2], $N/\xi \approx 2\tau$ stays bounded, while here it grows like $2\log N$. By part (b), the boundary condition moves the coupling at the crossing by an amount that decays only like $1/\log N$.

Stepanyan et al. [6, Section III.B] call the periodic central crossing the second thermal transition of the finite chain. They give the approximate formula $\beta J \approx (2\log N - 1)/5$ [6, Eq. (16)], and their figure 1(b) shows the fitted line $\beta J \approx 0.3864 \log N - 0.21055$. Table 2 and Figure 6 compare both with the exact crossing and with (37) without its error term. The line is accurate for $10 \le N \le 100$, but neither approximation has a $\log\log N$ term. The main terms of (37) have slope $\frac{1}{2} - 1/(2\log N)$ in $\log N$, up to $O(\log\log N/(\log N)^2)$, so the exact curve bends slowly toward slope $\frac{1}{2}$. The secant slope of the exact periodic coupling is 0.385 on $10 \le N \le 10^2$, close to the fitted 0.3864, then 0.417 on $10^2 \le N \le 10^3$, and 0.450 on $10^4 \le N \le 10^5$. At the tabulated sizes $N = 10^3, 10^4, 10^5, 10^6$, the main terms of (37) are closer to the exact value than either approximation, and at $N = 10^5$ they are off by 0.002, against 0.171 and 0.338.

Remark 5.6 (other comparisons). We compare one central bin with one endpoint, as Stepanyan et al. [6] do. Comparing instead with both endpoints, or more generally solving $S^X_{N,m_N}(u) = \Theta$ for a fixed $\Theta > 0$, only adds $\log\Theta$ to the constant c_X. For periodic ends, Step 2 of the proof of Theorem 5.2 goes through with $\log\Theta$ added to the right side of its crossing equation. For free ends the same argument works with Theorem 3.3(i) and (9) in place of Theorem 5.1, and with $\frac{1}{8}$ in place of $\frac{3}{8}$. So (35) and (11) hold for these roots with $c_X + \log\Theta$ in place of c_X, and the coupling moves by $-(\log\Theta)/(2\log N)$ to leading order. For $\Theta = 2$ the free constant becomes $c_0 + \log 2 = c_{\mathrm{P}}$, so this $\log 2$, which comes from the convention, should not be confused with the shift by $\log 2$ between the boundary conditions in Theorem 5.2. For odd N, adding the two central bins as well as the two endpoints gives $\Theta = 1$ again. The bracket of Theorem 3.4 is proved only for $\Theta = 1$.

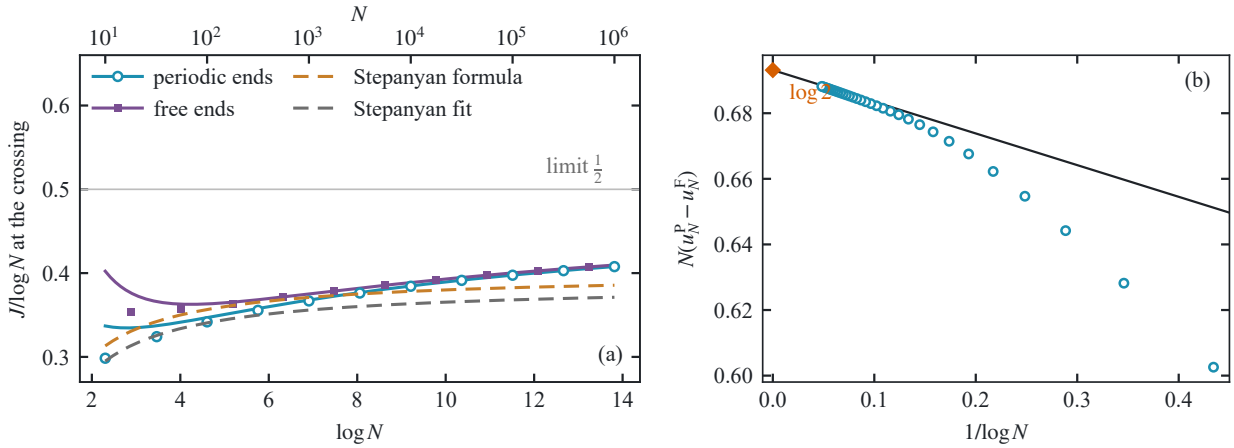


Figure 6: (a) The coupling at the central crossing divided by $\log N$, against $\log N$ for $10 \le N \le 10^6$, from the exact run polynomials for periodic (circles) and free (squares) ends, with the main terms of (37) and of Corollary 5.5(a) (solid lines of the same color) and the two approximations of Stepanyan et al. [6] (dashed). The thin line is the limit $\frac{1}{2}$. The two approximations have no $\log\log N$ term and a slope in $\log N$ below $\frac{1}{2}$, so they drift away. (b) The boundary shift $N(u_N^{\mathrm{P}} - u_N^{\mathrm{F}})$ for $10 \le N \le 10^9$, from the exact run polynomials, against $1/\log N$, with the first two terms $\log 2 + (\frac{1}{4} - \frac{1}{2}\log 2)/\log N$ of (36) (line). The shift tends to $\log 2$ (diamond), and at $N = 10^9$ it differs from the two terms by 3×10^{-4}.

5.4 Domain walls and the energy gap

We now describe the crossing in terms of domain walls. Given that the free chain is in bin m_N, a configuration with j domain walls has probability proportional to u^j, so the number j of domain walls has the law $W(N, m_N, j)u^j/S_N(u)$. Let $\mathcal{W}_N(u) = uS_N'(u)/S_N(u)$ be its mean, and put $\chi(w) = I_0(w)/(I_0(w) + I_1(w))$.

Proposition 5.7 (domain walls at the crossing)**.** *Let $N \ge 2$ and $u > 0$, put $w = Nu$, and suppose that $w(w+2) < 2N$. Let $\Delta = -\log(1 - w(w+2)/(2N))$. Then for every $\delta > 0$,*

$$\bigl|\mathcal{W}_N(u) - w - \chi(w)\bigr| \le w(e^\delta - 1) + \frac{\Delta}{\delta}.$$

At the crossing, as $N \to \infty$, $\mathcal{W}_N(u_N) = z_N + \frac{1}{2} + \frac{1}{8z_N} + O\bigl((\log N)^{-2}\bigr)$.

So the central bin at the crossing carries about half a domain wall more than the switching scale z_N. Numerically, $\mathcal{W}_N(u_N) - z_N$ is 0.5191, 0.5166 and 0.5132 at $N = 10^3, 10^4, 10^5$, against 0.5224, 0.5162 and 0.5126 for $\frac{1}{2} + \frac{1}{8z_N}$.

Proof. Put $\vartheta = \log u$, $x = hu = w/2$ with $h = N/2$, $g(\vartheta) = \log S_N(e^\vartheta)$ and $G(\vartheta) = \log F(he^\vartheta)$. Both are logarithms of power series in e^ϑ with nonnegative coefficients, so both are convex in ϑ, and $g'(\vartheta) = \mathcal{W}_N(u)$. By Theorem 3.3(i), $G(\vartheta) - \log h - \Delta \le g(\vartheta) \le G(\vartheta) - \log h$, and the upper bound holds at every ϑ. From $F' = 2F + 2I_0(2x)$ we get $G'(\vartheta) = xF'(x)/F(x) = w + \chi(w)$. The function χ decreases, since I_1/I_0 is the mean of $\cos t$ under the density proportional to $e^{w\cos t}$ on $[0, \pi]$ [12, Eq. 10.32.3], and the derivative of I_1/I_0 in w is the variance of $\cos t$, which is positive. By convexity,

$$\mathcal{W}_N(u) \le \frac{g(\vartheta+\delta) - g(\vartheta)}{\delta} \le \frac{G(\vartheta+\delta) - G(\vartheta) + \Delta}{\delta} \le G'(\vartheta+\delta) + \frac{\Delta}{\delta} \le we^\delta + \chi(w) + \frac{\Delta}{\delta},$$

and in the same way $\mathcal{W}_N(u) \geq we^{-\delta} + \chi(w) - \Delta/\delta$. Since $1 - e^{-\delta} \leq e^{\delta} - 1$, this proves the bound.

At the crossing $w = Nu_N = z_N + O(L^2/N)$ and $\Delta = O(L^2/N)$, and $\delta = (L/N)^{1/2}$ makes the bound $O(L^{3/2}N^{-1/2})$. The large-argument forms of I_0 and I_1 give $\chi(w) = \frac{1}{2} + \frac{1}{8w} + O(w^{-2})$, which proves the second claim. □

Since $u = e^{-2J}$, the definition of $\mathcal{W}_N$ gives

$$\frac{d}{dJ} \log \frac{P_N(m_N)}{P_N(0)} = -2\mathcal{W}_N(u).$$

The energy $-\sum_{i<N} \sigma_i\sigma_{i+1}$ of a configuration exceeds that of the all-up configuration by twice its number of domain walls. So at the crossing the mean energy gap between the central bin and the all-up configuration is $2\mathcal{W}_N(u_N) = 2\log N - \log\log N + O(1)$, by Proposition 5.7 and Corollary 3.7(b). Over a range of couplings of width $1/(2\mathcal{W}_N(u_N))$, which is 0.061 at $N = 10^4$ and 0.048 at $N = 10^5$, the ratio of the central bin to an endpoint changes by about a factor e. The same proof with Theorem 5.1 in place of Theorem 3.3, where G becomes $\log(xI_1(2x))$ and $G' = wI_0(w)/I_1(w)$, gives the mean number $Nu_N^{\mathrm{P}} + \frac{1}{2} + O(1/\log N)$ of domain walls in the central bin at the periodic crossing, for even N.

The count of about $\log N$ walls also accounts for a second fit of Stepanyan et al. They measure the mean domain length $l_d = \langle N/(n_d + \delta_{0,n_d})\rangle$, with n_d the number of domain walls of the periodic chain, at their second transition and fit $l_d \propto N^{0.757}$ on $20 \leq N \leq 100$ [6, Eq. (18)]. At the coupling J_N^{P} of the exact periodic crossing, where in the whole chain $n_d = 2r$ has weight proportional to $\binom{N}{2r}u^{2r}$, we get $l_d = 9.27$ at $N = 20$ and $l_d = 31.21$ at $N = 100$, and a least-squares exponent 0.755 over the even N in that range. With about $\log N$ walls one expects l_d of order $N/\log N$, whose least-squares exponent on that range is 0.74 and tends to 1 only slowly. So the fitted power is close to a local slope of $N/\log N$, just as the slope 0.3864 of their fitted coupling is close to the secant slope 0.385 of the exact coupling on $10 \leq N \leq 10^2$.

Remark 5.8 (full order against one bin). The chain is fully ordered, all up or all down, with probability $P_N(0) + P_N(N) = p^{N-1}$. This is $\frac{1}{2}$ at $u_{\mathrm{ord}} = 2^{1/(N-1)} - 1$, that is, at the coupling $J_{\mathrm{ord}} = \frac{1}{2}\log\frac{N-1}{\log 2} + O(1/N)$. By (2), $J_N^{\mathrm{F}} = \frac{1}{2}\log(N/z_N - 1)$, so

$$J_{\mathrm{ord}} - J_N^{\mathrm{F}} = \tfrac{1}{2}\log\frac{z_N}{\log 2} + O\Big(\frac{\log N}{N}\Big) = \tfrac{1}{2}\log\log N - \tfrac{1}{2}\log\log 2 + o(1).$$

The difference is 1.046 at $N = 10^3$ and 1.330 at $N = 10^5$. So full order wins against the rest of the chain only at a coupling about $\frac{1}{2}\log\log N$ above the central crossing. The $\log\log N$ in J_N^{F} measures how much less likely one central bin is than the whole disordered band.

5.5 A magnetic field

A dimensionless field $\mathcal{B} = \beta H > 0$ (βh in the notation of [6]) multiplies the weight of a configuration by $e^{\mathcal{B}(\sigma_1+\cdots+\sigma_N)}$. Magnetization $N - 2a$, with a spins down, then has weight $e^{-2\mathcal{B}a}$ relative to full magnetization, beyond the zero-field ratio. By the symmetry of the row, its ratio to the all-up configuration is

$$R_{N,a}^X(u) = S_{N,a}^X(u)\, e^{-2\mathcal{B}a} \qquad (1 \leq a \leq N-1),$$

for either boundary $X \in \{\mathrm{F}, \mathrm{P}\}$. We keep $\mathcal{B}$ fixed as N grows and as the coupling varies (this is different from varying the temperature at fixed physical field H). The ratio increases strictly from 0 to ∞ in u, so it equals 1 at a single $u = u_N^{(a)}$, and the all-up configuration is more likely than

magnetization $N-2a$ exactly when $u < u_N^{(a)}$. The roots depend on X and $\mathcal{B}$, which we suppress in the notation.

There are two comparisons behind the transitions of Stepanyan et al. [6, Section III.C]. The one-spin crossing is $u_N^{(1)}$. For $N \geq 3$, define

$$u_N^{(\geq 2)} = \min_{2 \leq a \leq N-1} u_N^{(a)}.$$

At this value the all-up configuration ties with the most likely bin having at least two down spins, excluding the all-down configuration (whose ratio to all up is always $e^{-2\mathcal{B}N} < 1$). The all-up configuration is strictly more likely than every other magnetization bin exactly when $u < \min\{u_N^{(1)}, u_N^{(\geq 2)}\}$. Even when there is only one transition in the sense of that paper, these two crossing values can differ.

Both comparisons come from the edge of the row. The right variable is $y = Nu^2$, which differs from the variable bu^2 of Section 4.4 by the factor $1 - a/N \to 1$. With H_a of (25), let $y_a(\mathcal{B})$ be the positive root of $H_a(y) = e^{2\mathcal{B}a}$. So $y_1(\mathcal{B}) = e^{2\mathcal{B}}$, and at $\mathcal{B} = 0$ these are the roots y_a of Section 4.4.

Theorem 5.9 (edge crossings in a fixed field)**.** *Fix a dimensionless field $\mathcal{B} > 0$ and either free or periodic ends. Let $y_a(\mathcal{B}) > 0$ solve $H_a(y) = e^{2\mathcal{B}a}$, with H_a as in (25).*

(a) For every fixed integer $a \geq 1$,

$$N(u_N^{(a)})^2 \longrightarrow y_a(\mathcal{B}) \qquad (N \to \infty).$$

In particular, $N(u_N^{(1)})^2 \to e^{2\mathcal{B}}$.

(b) As $a \to \infty$, $y_a(\mathcal{B}) \to \infty$, so the minimum

$$y_*(\mathcal{B}) = \min_{a \geq 2} y_a(\mathcal{B})$$

is attained, and

$$N(u_N^{(\geq 2)})^2 \longrightarrow y_*(\mathcal{B}) \qquad (N \to \infty).$$

(c) With $\mathcal{B}_c = \frac{1}{2}\log 2$,

$$\begin{cases} y_*(\mathcal{B}) < e^{2\mathcal{B}}, & 0 < \mathcal{B} < \mathcal{B}_c, \\ y_*(\mathcal{B}) = e^{2\mathcal{B}} = 2, & \mathcal{B} = \mathcal{B}_c, \\ y_*(\mathcal{B}) > e^{2\mathcal{B}}, & \mathcal{B} > \mathcal{B}_c. \end{cases}$$

In terms of the coupling, these two crossings are

$$J_N^{(1)} = \tfrac{1}{4}\log N - \tfrac{1}{2}\mathcal{B} + o(1), \qquad J_N^{(\geq 2)} = \tfrac{1}{4}\log N - \tfrac{1}{4}\log y_*(\mathcal{B}) + o(1).$$

The field moves the last crossing with all up to the edge scale $J = \frac{1}{4}\log N + O(1)$ (in zero field it comes from the middle, with leading coupling $\frac{1}{2}\log N$). For large N, if $\mathcal{B} < \mathcal{B}_c$, another bin is still more likely when the one-spin bin ties with all up. So all up becomes most likely at a later crossing, separated from the one-spin crossing by a positive limiting gap in J. If $\mathcal{B} > \mathcal{B}_c$, the one-spin bin is last to fall below all up. This gives only one transition to an all-up maximum, though the two crossing values defined above are still different. At $\mathcal{B} = \mathcal{B}_c$, their coupling gap tends to zero as $N \to \infty$ (the two crossings need not agree at finite N).

This recovers the limiting dimensionless threshold in Stepanyan et al. [6, Eq. (20)]. Their physical threshold is $h_c \approx 2J_{\text{ex}}\log 2/\log(N/2)$, where J_{ex} denotes their exchange coupling. At the merging point $\beta J_{\text{ex}} \sim \frac{1}{4}\log N$, so $\beta h_c \to \frac{1}{2}\log 2$. Theorem 5.9 gives the same limit for their periodic chain, and also for free ends. The agreement comes from the common fixed-edge limit H_a proved below.

Proof. (a) For free ends we use (28). With $b = N - a$ and $u = \sqrt{y/N}$, the edge variable $bu^2 = y(1 - a/N)$ tends to y, $V_r = U_r(b) \to 1$ for each $r \le a$, and the first two sums carry the factors $2\sqrt{y/N}$ and y/N, so $S^{\mathrm{F}}_{N,a}(u) \to H_a(y)$ uniformly for y in compact sets. For periodic ends, in (33) the term with r runs is $\frac{N}{r}\binom{a-1}{r-1}\binom{b-1}{r-1}u^{2r}$, and $\binom{b-1}{r-1}u^{2r-2} \to y^{r-1}/(r-1)!$ uniformly in the same sense, so $S^{\mathrm{P}}_{N,a}(u) \to \sum_{r=1}^{a}\binom{a-1}{r-1}y^r/r! = H_a(y)$. The limit $H_a(y)e^{-2\mathcal{B}a}$ increases strictly and continuously in y from 0 to ∞, and $R^X_{N,a}$ increases in u, so the roots converge.

(b) We need a bound uniform in a. Let $Y > 0$ and $0 < y \le Y$. By Theorems 4.3 and 5.1 the ratio is at most its Bessel form, and $I_0, I_1 \le e^z$. For free ends with $s = \sqrt{a(N-a)}$ and $c = N/(2s)$ this gives $S^{\mathrm{F}}_{N,a}(u) \le 2u(1+c)e^{2su}$, and for periodic ends $S^{\mathrm{P}}_{N,a}(u) \le (Nu/s)e^{2su}$. For every interior bin, $a(N-a) \ge N-1$, so $Nu/s \le \sqrt{2y}$. Also $2su \le 2\sqrt{ya}$ and $2u \le 2\sqrt{y}$. Thus for either boundary and every $1 \le a \le N-1$,

$$R^X_{N,a}(u) \le 4\sqrt{Y}\, e^{2\sqrt{Ya}-2\mathcal{B}a},$$

which is below 1 for every $a \ge a_Y$, uniformly in N, for some a_Y that depends only on Y and $\mathcal{B}$. In particular, every such interior bin has $N(u^{(a)}_N)^2 > Y$. Letting $N \to \infty$ in the displayed bound with a fixed, and using (a), gives $H_a(y)e^{-2\mathcal{B}a} \le 4\sqrt{Y}\, e^{2\sqrt{Ya}-2\mathcal{B}a} < 1$ for $a \ge a_Y$ and $y \le Y$, so $y_a(\mathcal{B}) > Y$. Hence $y_a \to \infty$ and the minimum is attained. Take $Y = y_*(\mathcal{B}) + 1$. For large N the minimum over $a \ge 2$ of $N(u^{(a)}_N)^2$ is a minimum over $2 \le a < a_Y$, and (a) gives its limit.

(c) Put $\lambda = e^{-2\mathcal{B}}$ and $y = e^{2\mathcal{B}} = 1/\lambda$. Then

$$H_a(y)e^{-2\mathcal{B}a} = \sum_{r=0}^{a-1}\binom{a-1}{r}\frac{\lambda^{a-1-r}}{(r+1)!} \le \sum_{r=0}^{a-1}\binom{a-1}{r}\lambda^{a-1-r}2^{-r} = \Big(\lambda + \frac{1}{2}\Big)^{a-1},$$

since $(r+1)! \ge 2^r$. If $\mathcal{B} > \frac{1}{2}\log 2$, then $\lambda < \frac{1}{2}$ and the right side is below 1 for every $a \ge 2$. As H_a increases, this gives $y_a(\mathcal{B}) > e^{2\mathcal{B}}$ for every $a \ge 2$, and so $y_*(\mathcal{B}) > e^{2\mathcal{B}}$ because the minimum is attained. If $\mathcal{B} < \frac{1}{2}\log 2$, then at $a = 2$ the sum is $\lambda + \frac{1}{2} > 1$, so $y_2(\mathcal{B}) < e^{2\mathcal{B}}$. At $\mathcal{B} = \frac{1}{2}\log 2$, the same bound gives $H_a(2)e^{-2\mathcal{B}a} \le 1$ for every $a \ge 2$, with equality at $a = 2$. Hence $y_a(\mathcal{B}) \ge 2$ and $y_*(\mathcal{B}) = y_2(\mathcal{B}) = 2$. □

The minimizing a grows as the field weakens in the following examples. For $\mathcal{B} = 0.1$, 0.2, 0.3 and 0.5, respectively, it is 11, 4, 2 and 2, and $y_* = 0.757$, 1.287, 1.764 and 2.972. At $N = 10^4$, in the same order of fields, $N(u^{(\ge 2)}_N)^2$ is 0.758, 1.288, 1.764 and 2.973 for periodic ends, and 0.741, 1.266, 1.738 and 2.939 for free ends. We checked the values at $N = 10^4$ directly from the full run polynomials. Interval bounds give the displayed digits and exclude competing bins near the edge, and the tail bound in the proof of Theorem 5.9(b) excludes all the remaining bins. The free chain converges more slowly because of the extra terms from its ends (of relative size $N^{-1/2}$). Here the field is fixed as N grows. If it decreases with N, the comparison should move back toward the middle of the row, and that crossover remains open.

6 Discussion

The crossing studied by Stepanyan et al. [6] compares the all-up configuration with one central magnetization bin. Around the crossing there is a band of magnetizations of width $N/\sqrt{\log N}$, with the parabolic free-energy profile of Remark 5.4. For free ends the central bin has about $\log N$ walls on average, giving the mean energy gap

$$2\mathcal{W}_N(u_N) = 2\log N - \log\log N + O(1)$$

above all up, in units of the exchange coupling (Proposition 5.7). Thus a factor e change in their probability ratio takes a coupling change of about $1/(2\mathcal{W}_N(u_N))$. This width decreases like $1/\log N$, whereas the usual $1/N$ rounding at a first-order transition comes from an energy gap proportional to the volume [25, 26, 27]. We therefore interpret this crossing as a finite-size crossover, consistent with the microcanonical analysis of finite chains in [28]. The $\log\log N$ term comes from balancing one wall-free configuration with the many configurations in a central bin. The $\log 2$ boundary shift comes from the restriction to even numbers of walls on a ring.

There are three relevant coupling scales. At the adjacent crossing, $\xi \approx \sqrt{N}/2$ (Proposition 4.2). For periodic ends this is the first crossing of [6], exactly at $J = \frac{1}{4}\log N$. At the central crossing, $\xi \approx N/(2\log N)$ (Corollary 5.5). For free ends, full order (all up or all down) has probability $\frac{1}{2}$ when $N/\xi \approx 2\log 2$ (Remark 5.8). Only at this last scale is the correlation length comparable with the chain length. Its coupling lies $\frac{1}{2}\log\log N + O(1)$ above the central crossing. This separation matters because the central crossing compares all up with just one bin. Full order must become more likely still before its weight equals that of the whole remaining chain.

The Bessel bounds behind these statements (Theorems 3.3, 4.3 and 5.1) hold at every N and every coupling. Through Corollary 4.4, Theorem 4.3 gives the fixed-magnetization partition function of the free chain used in [3, 5] with an explicit relative error. The error stays small when the mean number of domain walls grows with N. The approach extends to the d-state Potts chain [11]. Three questions are open. First, the bracket of Theorem 3.4 is proved only for free ends and for the level $\Theta = 1$ of Remark 5.6, and the same method should give it for periodic ends and for other levels. Second, a field that shrinks with N should move the comparison from the edge back toward the middle of the row, and we do not treat it. Third, Theorem 5.9 has no rate of convergence.

7 Verification

Remark 3.9 is computer-assisted. Otherwise the proofs use two computations. The first consists of the integer certificates for $2 \le N \le 174$, in Step 4 of the proof of Theorem 3.4 and in Proposition 3.8. The second is the evaluation at $N = 2774$ in Corollary 3.7(d). Both are described where they are used. Separate Python scripts check the exact law, the orderings, the Bessel bounds, the asymptotic formulas and the couplings of Table 2. Only one of them needs a library outside the standard one. Where possible a number is computed in two ways. For example, the couplings of Table 2 come from the run polynomials and again from the transfer matrix of the chain (using NumPy), and the two agree to 10^{-12}.

Every numbered result of the paper is formalized in Lean 4, in the folder `kagey-problems/problem131/lean/PaperA` of the repository named in the Data availability statement, except the Mehler–Heine limit for real parameters in Remark 4.7, which we cite (Lean proves it for nonnegative integer parameters). The results stated at every N, among them Theorem 3.4 and Proposition 3.8 for all $N \ge 2$, are proved as stated. Each asymptotic error term is replaced by a bound that holds from some point on and is at least as strong as the paper's, with explicit constants in Corollary 3.7(b) and (c), Corollary 5.5 and Proposition 5.7. For $2 \le N \le 174$ the Lean kernel checks the same certificates as Step 4 of the proof of Theorem 3.4, with rational bounds on ζ_N that Lean proves itself, and for Proposition 3.8 it checks rational certificates of its own on the same brackets. For Corollary 3.7(d) Lean proves its own rational bounds on ζ_{2774} and extends the inequality to every larger N, and for Remark 3.9 it checks one rational certificate between z_N and ζ_N for each $2 \le N \le 2773$. Lean also proves the finite Ising dictionary and the large-argument forms of I_0 and I_1 of Section 1.2, the correlations $\langle\sigma_0\sigma_r\rangle = e^{-r/\xi(J)}$ of the infinite chain (with

$\langle \sigma_i \sigma_{i+r} \rangle = \tanh^r J$ in every finite free chain), the consequences displayed after Theorem 4.9, and that $R^X_{N,a}$ is the ratio of the Ising weights in the field of the magnetizations $N-2a$ and N. With kernel-checked rational bounds it checks that every value printed in Tables 1 and 2 is the correct rounding of the quantity it names, and it bounds most decimals quoted in the text and the captions. Not formalized are the couplings $J_N^{(1)}$ and $J_N^{(\geq 2)}$ displayed after Theorem 5.9, the mean number of domain walls at the periodic crossing stated after Proposition 5.7, the Lambert form of ζ_N, and the numbers 0.755, 0.74 and 0.013 of Sections 5.4 and 4.5. Only the standard axioms are used. The file `MAP.md` in that folder gives the Lean theorem for each result, or says that there is none.

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I thank Peter Kagey, whose Open Problem Collection [29] asks for this crossing for a Galton board whose ball keeps its direction with probability p.

Use of AI tools. I used Claude Opus 5.5 (Anthropic, through Claude Code, in September and October 2026) and GPT-6 Astra (OpenAI, through Codex) in this work. I had conjectured a bracket for the crossing at every N, but it had gaps. Claude Opus 5.5 proposed an improved version, with a proof that uses the Bessel bounds of Lemma 3.6, and I then wrote my own version, which is Theorem 3.4 as stated here. Claude Opus 5.5 also proposed Corollaries 4.4 and 5.5 and Proposition 5.7, and drafted their proofs, which I rewrote by hand. The other results and their proofs are my own. I used both models to check and revise the paper, to search the literature and draft some of the comparisons with it, and to write parts of the Python verification scripts and the figures.

The Lean 4 formalization in `kagey-problems/problem131/lean/PaperA` was generated using Claude Opus 5.5 through conversation, and it was revised and checked against the results by me (`MAP.md` in that folder gives the Lean theorem for each result). Every proof, in the form present in this paper, was written by me, and all mathematical claims, proofs and computations in this paper are my responsibility.

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Data availability statement

The code and data are in the folder `kagey-problems/problem131` of https://github.com/apemm/Kagey-Problems, in the release `paperA-arxiv-v3`. Run `python -B verify_all.py --paper a` in that folder to repeat the checks. `SUPPLEMENT.md` describes each script, and `paperA/COMPUTATION.md` gives the commands and dependencies. The certificates of Theorem 3.4 are in `paperA/data/every_n_certificates.csv`. The script `paperA/verify_ising_coupling.py` prints Table 2. Numbers in Sections 3 and 5 that no script prints are in `paperA/data/quoted_values.txt`.

A Proofs of the bracket and of the crossing equation

This appendix proves Lemma 3.6, the analytic half of Theorem 3.4, the identity behind the certificates and Proposition 3.8. The letters s, v, y, λ_k, $g_\pm$, Y, Y', $\bar{w}$, $\bar{w}'$, r, r', t, t', τ, κ, m, B, D,

β, Δ, δ and ϕ are local to this appendix, and each is defined where it first shows up. The letter t is an angle in Appendix A.1 and a bound in Appendices A.2 and A.4. We round every decimal constant in the safe direction, namely so that the inequality it enters stays true.

A.1 Proof of Lemma 3.6

For integer n, $I_n(z) = \frac{1}{\pi}\int_0^\pi e^{z\cos t}\cos(nt)\,dt$ [12, Eq. 10.32.3], so $I_0(z)+I_1(z) = \frac{1}{\pi}\int_0^\pi e^{z\cos t}(1+\cos t)\,dt$. Substitute $s = 1-\cos t$, so that $\sin t = \sqrt{s(2-s)}$ and $1+\cos t = 2-s$, and then $s = v/z$. This gives

$$\mathcal{E}(z) = \frac{1}{\sqrt{\pi}}\int_0^{2z} e^{-v}v^{-1/2}\sqrt{1-\frac{v}{2z}}\,dv. \tag{40}$$

On $[0,1]$ we have $\sqrt{1-y} = 1-\sum_{k\ge1}\lambda_k y^k$ with every $\lambda_k > 0$, $\lambda_1 = \frac12$ and $\lambda_2 = \frac18$, and $\sum_k \lambda_k = 1$ by monotone convergence as $y\uparrow 1$. So $\sum_{k\ge3}\lambda_k = \frac38$, and $y^k \le y^3$ gives $\sum_{k\ge3}\lambda_k y^k \le \frac38 y^3$. Hence, for $0\le y\le 1$,

$$g_-(y) := 1-\frac{y}{2}-\frac{y^2}{8}-\frac{3y^3}{8} \le \sqrt{1-y} \le 1-\frac{y}{2}-\frac{y^2}{8} \le g_+(y) := 1-\frac{y}{2}-\frac{y^2}{8}+\frac{y^3}{8}.$$

The cubic g_- decreases and $g_-(1) = 0$, so $g_- \le 0$ on $[1,\infty)$. Also $8g_+(y) = (y-2)^2(y+1)+2(y-1)^2+2 > 0$ for $y \ge -1$. Put $y = v/(2z)$ in (40), replace $\sqrt{1-y}$ by $g_\pm$ and extend the integral to ∞. The added piece is at most 0 with g_- and at least 0 with g_+, so the lower bound survives with g_- and the upper with g_+. Since $\frac{1}{\sqrt{\pi}}\int_0^\infty e^{-v}v^{k-1/2}\,dv = \Gamma(k+\frac12)/\Gamma(\frac12)$ equals $1, \frac12, \frac34, \frac{15}{8}$ for $k = 0,1,2,3$, the two bounds integrate to the two sides of the lemma. The bound $\mathcal{E} \le 1$ follows in the same way from $\sqrt{1-y} \le 1$. Finally $-\frac{3}{128z^2}+\frac{15}{512z^3} \le 0$ exactly when $z \ge \frac54$.

A.2 Proof of Theorem 3.4 for $N \ge 175$

Step 1. Bounds that hold for every $\zeta \ge 4$. Fix $N \ge 175$ and write $\zeta = \zeta_N$, $Y = \zeta + \frac{1}{8\zeta}$, $r = Y/N$ and $n(\zeta) = \sqrt{8\zeta/\pi}\,e^\zeta$. Then $n(\zeta_N) = N$ by (8), and $\zeta \ge 4$ since $n(4) < 175$. For $k = 1,2,3$ the function $Y^k/n(\zeta)$ decreases on $[4,\infty)$, because its logarithmic derivative is $k(1-\frac{1}{8\zeta^2})/Y - 1 - \frac{1}{2\zeta} < \frac{k}{Y} - 1 < 0$. The same holds for Y^4/n, since $Y > 4$, and for ζ/n and ζ^2/n. The values at $\zeta = 4$ give, for every $\zeta \ge 4$,

$$r \le 0.023135, \quad \frac{Y^2}{N} \le 0.093262, \quad \frac{Y^3}{N} \le 0.375960, \quad \frac{Y^4}{N} \le 1.51559, \quad \frac{\zeta(\zeta+1)}{N} \le 0.114777.$$

Step 2. The upper bound. Let $\bar w = Y/(1-r)$, so that $N\bar w/(N+\bar w) = Y$, and let $\delta = \bar w(\bar w+2)/(2N)$. By Lemma 3.5(b) and (9) it suffices to show that $\mathcal{G} := \psi(\bar w)-\psi(\zeta)+\log\mathcal{E}(\bar w)+\log(1-\delta) > 0$. We bound the three parts from below. First, $\psi(\bar w)-\psi(\zeta) = (\bar w - Y)+(Y-\zeta)+\frac12\log\frac{Y}{\zeta}+\frac12\log\frac{\bar w}{Y}$, where $\bar w - Y = \frac{Yr}{1-r}$, $Y-\zeta = \frac{1}{8\zeta}$, $\log\frac{Y}{\zeta} = \log(1+\frac{1}{8\zeta^2}) \ge \frac{1}{8\zeta^2}-\frac{1}{128\zeta^4}$ and $\log\frac{\bar w}{Y} = -\log(1-r) \ge r$. Second, Lemma 3.6 gives $\mathcal{E}(\bar w) \ge 1-t$ with $t = \frac{1}{8\bar w}+\frac{3}{128\bar w^2}+\frac{45}{512\bar w^3}$. Since $\bar w \ge Y \ge \zeta$ and $\frac{1}{8Y} = \frac{1}{8\zeta}-\frac{1}{8\zeta(8\zeta^2+1)} \le \frac{1}{8\zeta}-\frac{1}{64\zeta^3}+\frac{1}{512\zeta^5}$, we get $t \le \tau := \frac{1}{8\zeta}+\frac{3}{128\zeta^2}+\frac{37}{512\zeta^3}+\frac{1}{512\zeta^5}$ and $\log\mathcal{E}(\bar w) \ge \log(1-\tau) \ge -\tau-\frac{\tau^2}{2(1-\tau)}$. Third, $\delta = \frac{Yr}{2(1-r)^2}+\frac{r}{1-r}$ and $\log(1-\delta) \ge -\delta-\frac{\delta^2}{2(1-\delta)}$. Adding the three parts gives $\mathcal{G} \ge B + D$, where B collects the terms in ζ alone and D the terms with a factor r,

$$B = \frac{5}{128\zeta^2}-\frac{37}{512\zeta^3}-\frac{1}{256\zeta^4}-\frac{1}{512\zeta^5}-\frac{\tau^2}{2(1-\tau)}, \qquad D = \frac{r}{2(1-r)}\Big(\frac{Y(1-2r)}{1-r}-1-r\Big)-\frac{\delta^2}{2(1-\delta)}.$$

Both $\zeta\tau$ and τ decrease in ζ, so $\zeta^2 B$ increases. At $\zeta = 4$ we have $\zeta\tau < 0.135384$, $\tau < 0.033846$ and $\zeta^2 B > 0.011236$. Hence $B > 0$. In D the bracket is at least $4.03125(1-0.023683)-1.023135 > 2.912$, so the first term of D is at least $1.456\,r$. Write $\delta = rm$ with $m = \frac{Y}{2(1-r)^2} + \frac{1}{1-r}$. Then $\delta \le 0.07255$ and $rm^2 = \frac{Y^3}{4N(1-r)^4} + \frac{Y^2}{N(1-r)^3} + \frac{r}{(1-r)^2} \le 0.2276$, so $\frac{\delta^2}{2(1-\delta)} = r \cdot \frac{rm^2}{2(1-\delta)} \le 0.1228\,r$. Hence $D \ge 1.33\,r > 0$. So $\mathcal{G} > 0$ and $z_N < Y$.

Step 3. The lower bound. Let $\varepsilon = \varepsilon_N$, $Y' = Y - \varepsilon$, $r' = Y'/N$ and $\bar{w}' = Y'/(1-r')$, so that $N\bar{w}'/(N+\bar{w}') = Y'$. By Step 1, $\varepsilon \le 0.114777 + \frac{1}{256} < 0.1187$, so $Y' \ge \zeta - 0.1187 \ge 3.88$ and $\bar{w}' \ge Y'$. By Lemma 3.5(a) and (9) it suffices to show that $\mathcal{G}' := \psi(\bar{w}') - \psi(\zeta) + \log \mathcal{E}(\bar{w}') < 0$. We bound the two parts from above. First, $\psi(\bar{w}') - \psi(\zeta) = (Y' - \zeta) + \frac{Y'r'}{1-r'} + \frac{1}{2}\log\frac{Y'}{\zeta} - \frac{1}{2}\log(1-r')$, where $Y' - \zeta = \frac{1}{8\zeta} - \varepsilon$, $\frac{1}{2}\log\frac{Y'}{\zeta} \le \frac{1}{2}(\frac{Y'}{\zeta} - 1) = \frac{1}{16\zeta^2} - \frac{\varepsilon}{2\zeta}$ and $-\frac{1}{2}\log(1-r') \le \frac{r'}{2(1-r')}$. The two terms in r' add up to $\frac{Y'^2 + Y'/2}{N(1-r')}$. Second, Lemma 3.6 gives $\mathcal{E}(\bar{w}') \le 1 - t'$ with $t' = \frac{1}{8\bar{w}'} + \frac{3}{128\bar{w}'^2} - \frac{15}{512\bar{w}'^3} \in (0,1)$, so $\log \mathcal{E}(\bar{w}') \le -t'$. Here $\frac{1}{8\bar{w}'} = \frac{1}{8Y'} - \frac{1}{8N}$. Write $Y' = \zeta(1+\kappa)$ with $\kappa = \frac{1}{8\zeta^2} - \frac{\varepsilon}{\zeta}$. Then $\frac{1}{Y'} \ge \frac{1-\kappa}{\zeta}$, so $-\frac{1}{8\bar{w}'} \le -\frac{1}{8\zeta} + \frac{1}{64\zeta^3} - \frac{\varepsilon}{8\zeta^2} + \frac{1}{8N}$. Next, $x \mapsto x/(1-x/N)$ increases and $Y' \le Y$, so $\bar{w}' \le Y/(1-r)$ and $\frac{1}{\bar{w}'^2} \ge \frac{(1-r)^2}{\zeta^2(1+\frac{1}{8\zeta^2})^2} \ge \frac{(1-2r)(1-\frac{1}{4\zeta^2})}{\zeta^2}$. Finally $\frac{15}{512\bar{w}'^3} \le \frac{15}{512Y'^3}$. Adding, and using $\varepsilon(1+\frac{1}{2\zeta}) = \frac{\zeta^2 + \frac{3}{2}\zeta + \frac{1}{2}}{N} + \frac{1}{16\zeta^2} + \frac{1}{32\zeta^3}$, we get $\mathcal{G}' \le \beta + \Delta - \frac{\varepsilon}{8\zeta^2}$ with

$$\beta = -\frac{3}{128\zeta^2}(1-2r)\Big(1 - \frac{1}{4\zeta^2}\Big) + \frac{15}{512Y'^3} - \frac{1}{64\zeta^3}, \qquad \Delta = \frac{1}{N}\Big(\frac{Y'^2 + Y'/2}{1-r'} + \frac{1}{8} - \zeta^2 - \frac{3}{2}\zeta - \frac{1}{2}\Big).$$

For $\beta < 0$ it suffices that $\zeta^2/Y'^3 \le \frac{4}{5}(1-2r)(1-\frac{1}{4\zeta^2})$. The left side is at most $\zeta^2/(\zeta - 0.12)^3 \le 16/3.88^3 < 0.274$, since this function decreases for $\zeta > 0.12$ (its logarithmic derivative is $\frac{2}{\zeta} - \frac{3}{\zeta - 0.12} < 0$ there). The right side is at least $0.8 \cdot 0.9537 \cdot 0.9843 > 0.75$. For $\Delta < 0$, note that $Y' \le Y$ and $r' \le r$, so

$$\frac{Y'^2 + Y'/2}{1-r'} \le Y^2 + \frac{Y}{2} + \frac{Y^3/N + Y^2/(2N)}{1-r} \le \zeta^2 + \frac{\zeta}{2} + 0.2667 + 0.4326,$$

using $Y^2 + \frac{Y}{2} = \zeta^2 + \frac{\zeta}{2} + \frac{1}{4} + \frac{1}{16\zeta} + \frac{1}{64\zeta^2}$. This is less than $\zeta^2 + \frac{3}{2}\zeta + \frac{3}{8}$ because $\zeta \ge 4$. Hence $\mathcal{G}' < 0$ and $z_N > Y'$.

The constants were recomputed in 40-digit arithmetic, and as a check outside the proof we evaluated $\zeta^2 B$, D, β and Δ on $\zeta \in [4, 60]$ with step 0.001.

A.3 The certificates

Let $c, d > 0$ be integers. Multiplying (6) by d^{N-1} gives

$$d^{N-1} S_N(c/d) = \sum_j W(N, m_N, j)\, c^j d^{N-1-j},$$

an integer. By Theorem 2.1 it is the sum of $d^{\mathrm{rep}(w)} c^{\mathrm{chg}(w)}$ over the words w in bin m_N, that is, the numerator $n_N(m_N)$ of Remark 2.3 with $(\alpha, \beta) = (d, c)$. The recursion (3) with the weights d, c in place of p, q and initial values 1 computes the same sums. So $S_N(c/d) < 1$ exactly when this integer is less than d^{N-1}. This is the identity used in Step 4 of the proof of Theorem 3.4.

A.4 Proof of Proposition 3.8

Step 1. An exact formula for R_N. Write $w = w_* = Nu_N$, $z = z_N$, $\Lambda_N = 2\Phi(w)/N$ and $\delta = w(w+2)/(2N)$. At $w = w_*$ the display in the proof of Lemma 3.5 gives $1 \le \Lambda_N \le 1/(1-\delta)$ when $\delta < 1$, and (9) gives $\psi(w) + \log \mathcal{E}(w) = L + c_0 + \log \Lambda_N$. Since $w - z = wz/N$, $w/z = 1 + w/N$ and $\frac{1}{8z} = \frac{1}{8w} + \frac{1}{8N}$, the remainder $R_N = \psi(z) - L - c_0 - \frac{1}{8z}$ is

$$R_N = \log \Lambda_N - \Big(\log \mathcal{E}(w) + \frac{1}{8w}\Big) - \frac{1}{8N} - \frac{wz}{N} - \frac{1}{2}\log\Big(1 + \frac{w}{N}\Big). \tag{41}$$

Step 2. The lower bound for $N \ge 175$. By the proof of Theorem 3.4, $3.88 \le z \le w \le Y/(1-r)$. Note that $\log \Lambda_N \ge 0$ and $-\log \mathcal{E}(w) \ge -\log(1 - \frac{1}{8w}) \ge \frac{1}{8w}$ by Lemma 3.6, since $w \ge \frac{5}{4}$. So $R_N \ge -\frac{wz + w/2 + 1/8}{N} = -\frac{z^2 + z/2}{N - z} - \frac{1}{8N}$. This is at least $-\frac{z^2+z}{N}$ as soon as $N \ge \frac{2z^2(z+1)}{z - 1/4}$. The right side increases in z and $z \le Y$, so it suffices that $Y^2/N \le \frac{Y - 1/4}{2(Y+1)}$. The left side is at most 0.0933 and the right side at least 0.3757.

Step 3. The upper bound for $N \ge 175$. Let $t = \frac{1}{8w} + \frac{3}{128w^2} + \frac{45}{512w^3} \le \frac{1.09504}{8w}$, which holds as $w \ge 3.88$. Lemma 3.6 and $-\log(1-t) \le t + \frac{t^2}{2(1-t)}$ give

$$-\log \mathcal{E}(w) - \frac{1}{8w} \le \frac{3}{128w^2} + \frac{45}{512w^3} + \frac{t^2}{2(1-t)} \le \frac{0.0559}{w^2} \le \frac{0.0559}{z^2}.$$

Using $\log \Lambda_N \le \delta + \frac{\delta^2}{2(1-\delta)}$, $\log(1+x) \ge x - \frac{x^2}{2}$ and $wz \ge w^2 - w^3/N$, the rest of (41) is at most $-\frac{(w-1/2)^2}{2N} + \frac{w^3 + w^2/4}{N^2} + \frac{\delta^2}{2(1-\delta)}$. By the bounds on r and Y^k/N in Step 1 of the previous proof, with $w \le Y/(1-r)$, we have $\frac{w^3 + w^2/4}{N} \le 0.4278$ and $N\frac{\delta^2}{2(1-\delta)} = \frac{(w^4 + 4w^3 + 4w^2)/(4N)}{2(1-\delta)} \le 0.4945$, while $\frac{(w-1/2)^2}{2} \ge 5.71$. So the rest is negative and $R_N \le 0.0559/z^2 < \frac{1}{16z^2}$.

Step 4. The form in L for $N \ge 175$. We have $z < Y = L + c_0 - \frac{1}{2}\log\zeta + \frac{1}{8\zeta} < L$. Also $z > \zeta - 0.1187 \ge \nu_N - 0.1187 \ge L/2$, because $L - \log L \ge 1.18$ for $L \ge \log 175$. Then $\frac{1}{16z^2} \le \frac{1}{4L^2}$ and $\frac{z^2+z}{N} \le \frac{L^2+L}{N}$.

Step 5. Short chains. Here $N \le 174$. The function $\phi(z) = \psi(z) - L - c_0 - \frac{1}{8z}$ increases in z, and $R_N = \phi(z_N)$, while $\frac{1}{16z^2}$ and $-\frac{z^2+z}{N}$ decrease. So it suffices to check $\phi(z(c'/d)) \le \frac{1}{16z(c'/d)^2}$ and $\phi(z(c/d)) \ge -\frac{z(c/d)^2 + z(c/d)}{N}$ with the certificates, together with $z(c/d) \ge L/2$ and, for $N \ge 3$, $z(c'/d) \le L$. The outward interval calculation described in Step 4 of the proof of Theorem 3.4 verifies every comparison. The gaps exceed 0.037 (upper) and 0.081 (lower), with the smallest at $N = 174$. For $N = 2$ we have $S_2(u) = 2u$, so $u_2 = \frac{1}{2}$ and $z_2 = \frac{2}{3}$, and $\frac{2}{3} < \log 2$ is equivalent to $e^2 < 8$, which holds since $e^2 < 2.72^2 = 7.3984$.

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